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FAIR2Adapt

FAIR2Adapt
Deliverable 6.1 – Stocktaking report on adaptation data and knowledge needs

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Abstract:	This deliverable presents a cross-case analysis of adaptation knowledge needs and stakeholder engagement processes across six diverse climate adaptation case studies within the FAIR2Adapt project. Through a series of structured workshops, interviews, and recurrent coordination meetings, the work identifies how information is produced, shared, and interpreted within each case, and where bottlenecks emerge in translating knowledge into decision-making. The analysis focuses on

	both the specific requirements of each case study—ranging from scientific modelling of environmental risks to national adaptation planning and community-oriented knowledge platforms—and the broader patterns that shape adaptation practice across contexts. A central finding is that while high-quality scientific and policy information exists in all cases, its usability depends on the extent to which it becomes discoverable, interpretable, and aligned with the needs, capacities, and incentives of the actors expected to use it. The FAIR principles provide a guiding framework for improving these information pathways, but their application requires attention to institutional realities, communication practices, and the different ways communities engage with climate knowledge. This deliverable synthesises these insights, documents the current state of stakeholder engagement in each case study, and outlines the opportunities and limitations of advancing FAIR-based digital infrastructures to strengthen climate adaptation planning.
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Terms and Abbreviations

ACRONYM	DESCRIPTION
AI	Artificial Intelligence
API	Application Programming Interface
CCA	Climate Change Adaptation
CCH	Climate Connectivity Hub
CCT	Climate Connectivity Taxonomy
CI/CD	Continuous Integration / Continuous Deployment
CMIP	Coupled Model Intercomparison Project
Copernicus CDS	Copernicus Climate Data Service
CRA	Climate Risk Assessment
CS	Case study
CSV	Comma-Separated Values
DCAT-AP	Data Catalog Vocabulary Application Profile
DGGS	Discrete Global Grid System

ACRONYM	DESCRIPTION
DOI	Digital Object Identifier
DSA	Radiation and Nuclear Safety Authority
EEA	European Environment Agency
ERDDAP	Environmental Research Division's Data Access Program
ETCCDI	Expert Team on Climate Change Detection and Indices
EU	European Union
FAIR	Findable, Accessible, Interoperable, Reusable
FIP	Fair Implementation Profile
FDO	FAIR Digital Objects
HPC	High-Performance Computing
HWDI	Heat Wave Duration Index
IAEA	International Atomic Energy Agency
IPMA	Instituto Português do Mar e da Atmosfera (The Portuguese Institute for Sea and Atmosphere)

ACRONYM	DESCRIPTION
KPI	Key Performance Indicator
LLM	Large Language Model
MetNO	Norwegian Meteorological Institute
MPA	Marine Protected Area
netCDF	Network Common Data Form
NGO	Non-Governmental Organization
NIVA	Norwegian Institute for Water Research
NorESM-OC	Norwegian Earth System Model - Ocean Component
NUTS	Nomenclature of Territorial Units for Statistics
OPeNDAP	Open-source Project for a Network Data Access Protocol
PDF	Portable Document Format
RCP	Representative Concentration Pathway
RiOMar	River-dominated Ocean Margin

ACRONYM	DESCRIPTION
SCORM	Sharable Content Object Reference Model
SKOS	Simple Knowledge Organization System
SMR	Small Modular Reactor
SMS	Short Message Service
THREDDS	Thematic Real-time Environmental Distributed Data Service
TXx	Maximum value of daily maximum temperature in a given period (usually a year or a season)
WP	Work Package
WPL	Work Package Leaders
Zarr	N-Array Data format

Executive Summary

Aim of the Deliverable

- To identify the adaptation knowledge needs of the six FAIR2Adapt case studies and outline their stakeholder engagement plans.
- To provide a consolidated view of how data, processes, and actors interact across cases, and where FAIR principles can improve the usability, accessibility, and impact of climate-adaptation information.

Conclusion of the Deliverable

- The case studies reveal two overarching challenges: conveying large, complex scientific information to diverse, non-technical users; and enabling human-centred processes—engagement, coordination, and communication—required for adaptation decisions.
- FAIRification can meaningfully support these processes by improving discoverability, interoperability, and reusability of both quantitative and qualitative adaptation knowledge.

Methodology

- A design-thinking framework was used to explore user needs and information flows, including four workshop exercises: stakeholder mapping, process mapping, dashboard co-design, and user-story elicitation.
- Workshops, regular case-study meetings, Figma boards, and documented user requirements were analysed to understand data journeys, bottlenecks, and engagement strategies across all cases.

Major Findings, Results, and Recommendations

- Adaptation bottlenecks emerge not only from technical gaps but from the critical stage where information must be translated into decisions and practical action.
- Case studies highlight recurring needs: accessible visualisation of large datasets, clear stakeholder roles, consistent metadata and vocabularies, improved inter-institutional coordination, and mechanisms for translating scientific outputs into practical decisions.
- The project should prioritise FAIR Digital Objects, shared vocabularies, simplified decision-support tools, and structured feedback loops between technical work packages and case-study stakeholders.

Shortcomings / Limitations Identified

- Fragmented governance structures and dependency on personal networks hinder end-to-end information flow.

- Significant difficulties remain in FAIRifying qualitative information such as policy documents and narrative knowledge.
- Models often operate on timeframes that do not match real-world decision windows.
- Some case studies face limited data accessibility, licensing constraints, or stakeholder fatigue.

1. Introduction

The climate crisis is increasingly threatening the livelihoods of humans across the globe. This year, the Planetary Health Check (2025) report states that not only one more critical threshold i.e. ocean acidification, has been breached, but also none of the existing threats has moved in a positive direction despite the existence of numerous examples of scalable technologies across sectors. At the same time, lots of information is freely and readily available like never before with many scientific teams around the globe trying to communicate their findings in novel ways to amplify the impact. Decision makers are challenged with increasing damages and threats, and local communities around the world are experiencing the effects of the climate crisis in their daily lives. Just to mention a few of the most visible challenges, fishing communities see their livelihoods impacted by climate change, heatwaves cause deaths and fires in both forested and urban areas, droughts cause civil unrest and threaten agricultural output and food security.

How can digital technology mediate the impact of the climate crisis? This document explores six case studies with the aim of identifying specific user needs, bottlenecks, and the possibilities and difficulties of addressing these needs with technology. Each case offers a different window into how information is produced, circulated, and used—or not used—in real decision-making settings. More specifically, this project examines how the FAIR principles can help structure and connect these adaptation processes, ensuring that the knowledge emerging from the six case studies becomes more discoverable, interoperable, and usable across contexts and scales. In other words, while the objective is to provide a detailed report for each of the CS, the findings of the workshops and the interaction of this role within the consortium produced a rich set of information about the applicability of information for adaptation processes that goes beyond the processes outlined in the document's objectives.

In that sense, this report does more than summarise adaptation knowledge needs and stakeholder engagement plans. It also reflects the difficulties we encountered while trying to trace how information actually moves through each case study. Some of these challenges fall outside the strict categories expected in the deliverable—knowledge needs, engagement plans, methods, timelines, and synergies—but they are essential for understanding the real-world applicability of the FAIR principles to adaptation policy. At its core, the main objective of this deliverable is to document the knowledge needs that shape adaptation decisions across the six case studies and to provide a clear, grounded account of how stakeholders work, what they require, and where bottlenecks arise. Building on internal dialogues, the workshops, and the lived experience of each case study team, this document aims to inform reviewers and future efforts by highlighting both the opportunities and the limitations of turning FAIR theory into practice for climate adaptation.

2. Methodology

The approach used in this deliverable combines a user-centred methodology with recurring engagement across the consortium to understand adaptation knowledge needs and the data journeys that shape how information is produced, shared, and used. The purpose was to identify the concrete requirements of stakeholders involved in each case study and to document the bottlenecks that influence the flow of adaptation knowledge.

A design-thinking framework guided the process. Rather than applying the full methodology, the project used selected components that help reveal user needs in complex, multi-stakeholder environments. This approach was chosen because it supports rapid elicitation of requirements, clarifies pain points in existing information flows, and helps prioritise where FAIRification can provide the greatest value. The theoretical background of design thinking and its phases is provided in Annex I.

2.1 Workshop Design and Structure

In June 2025, WP (Work Package) 5 and WP6 conducted a series of three-hour online workshops with each of the six case studies. These workshops followed a standardised structure designed to generate comparable outputs across cases:

1. Stakeholder Mapping
Participants used metaphor-based archetype cards to identify the actors involved in each case study, discuss their roles, and surface hidden needs or dependencies in existing information flows. This exercise provided a shared understanding of the stakeholder ecosystem and where responsibilities, influence, or gaps may lie.

2. Process and Timeline Mapping
Groups constructed timelines representing the full decision process from data generation to action. Using visual prompts, participants identified key decision points, bottlenecks, missing datasets, and critical moments where information fails to move to the actors who need it. This helped clarify the challenges faced across case studies.

3. Dashboard Co-design
Participants sketched the features of an ideal dashboard or decision-support interface tailored to their user personas. This elicited practical requirements for usability, essential indicators, metadata, and visualisation needs.

4. Metadata and User Story Elicitation
Participants formulated user stories (“As a [role], I want [feature], so that [benefit]”), which provided structured, actionable requirements for FAIR Digital Objects and future tools. These inputs directly inform WP5 and WP3 activities.

All workshop interactions were documented using shared FIGMA boards, with one board per case study. Exercises 1–3 supported both Deliverable 6.1 and 5.1, while exercise 4 (‘Metadata and User Story Elicitation’) was primarily used for Deliverable 5.1.

2.2 Data Collection and Documentation

Information from the workshops and ongoing case-study meetings was captured through:

- Recorded sessions (when permitted), transcribed for accuracy
- Detailed notes on stakeholder narratives, decision processes, and identified bottlenecks
- FIGMA outputs including maps, diagrams, timelines, and user-generated metadata
- Consolidated lists of user needs, expected outputs, and risks for each case study

2.3 Focus of the Deliverable

While the workshops produced extensive information, this deliverable focuses on a subset directly relevant to adaptation knowledge needs and stakeholder engagement plans. The emphasis is on:

- The mapping of stakeholders and their interactions
- Identification of bottlenecks, information gaps, and user priorities
- How each case study has evolved during the first project year
- The implications of these findings for FAIRification processes and future work packages

The detailed workshop materials, full methodological reasoning, and expanded descriptions of design-thinking components are provided in Annex 1. The main document presents a synthesised view aimed at supporting reviewers’ understanding of how each case study’s needs were identified and how these will inform the next steps in FAIR2Adapt.

3. Case Study Reports

The six case studies presented in this deliverable represent distinct adaptation contexts, each with its own actors, knowledge needs, and degree of maturity. Some began the project with established relationships and ongoing engagement processes, while others used the first year to initiate new contacts and refine their scope. Their approaches range from highly quantitative, model-driven work to qualitative, policy-focused processes centred on institutional coordination and capacity building. Together, these case studies illustrate the diversity of adaptation challenges across the consortium and provide a grounded picture of where FAIR practices can meaningfully strengthen decision-making.

3.1 Case Study 1 (NERSC): Spread of radioactive isotopes in the Arctic under different climate scenarios to understand the climate change impact on radionuclide distribution for public and environmental safety.

Purpose:

1. Analyse climate predictions generated by the Norwegian Earth System Model NorESM under predefined scenarios in the Arctic

3.1.1 Problem and Decision Context

Large amounts of radionuclides are currently locked in permafrost. As the climate warms, melting permafrost could release these substances into ecosystems, posing risks to wildlife and human local communities. In simpler terms, materials that have been frozen and safely buried for centuries could suddenly leak into rivers, soils, and food chains once the ground thaws. There is still time to prepare for and mitigate these potential impacts, but only if we understand where the risks are, how quickly they may unfold, and who will be affected first.

Rapid permafrost melt caused by climate change could trigger sudden releases of contaminants, including radionuclides, threatening ecosystems and livelihoods. The project aims to improve the understanding of these impacts and strengthen connections between science and stakeholders.

Informing fisheries and local communities about radionuclide contamination risks involves multiple stakeholders. AMAP plays a central role, providing information to public officials and civil society.

3.1.2 Actors

The workshop identified several categories of stakeholders who interact with or depend on the information generated in this case study (Table A1). Although the stakeholder mapping offers an initial overview, additional actors and roles exist that are not yet fully captured. Before any data pipelines or visualisation tools can be implemented, further work is needed to clarify and prioritise the specific requirements of these stakeholder groups. This includes understanding how different users interpret scientific information, what formats they can work with, and what kinds of decisions they must support.

Knowledge custodians—primarily scientific and technical teams—are responsible for modelling radionuclide contamination and ensuring the accuracy of the underlying data. Their challenges include transferring large, complex datasets to non-technical audiences and operating within stringent regulatory frameworks during radiological events. The former will be addressed within this case study; the latter lies outside the project's scope. Platform owners, who aggregate and publish information through computational tools or web portals, play an essential bridging role. End-users include public officials, industry sectors such as fisheries and aquaculture, vendors of nuclear technology, media actors, and civil society groups. Each has distinct information needs that require tailored insights, needs that exceed the resources available within the current project and therefore must be prioritised carefully.

3.1.3 Priority Knowledge Needs and Bottlenecks

This case study demonstrates strong stakeholder engagement and effective knowledge translation. Scientific models and data are well maintained, but access for end-users outside academia are less well developed. The main challenge is enabling diverse stakeholders to access curated, visualized information tailored to their needs—for example, segmenting datasets by geography and providing tools for non-specialist audiences.

A critical concern is that climate change may trigger radionuclide release from thawing permafrost. Because the scale, frequency, and location of such events are uncertain, an end-to-end process for rapid information flow is essential. Current systems function as a two-step pipeline, but bottlenecks remain: (1) event detection and (2) tailoring and visualizing large datasets for specific stakeholders. Developing FAIR-compliant artifacts could help address these gaps.

3.1.4 Engagement and Methods

Stakeholder engagement in this case study occurs in two ways. First, through collaboration with scientific and technical partners and by interacting with the AMAP Radioactivity Expert Group. These partnerships have strong information-sharing frameworks, and clear data submission guidelines.

The second type of engagement involves a broader group—media, civil society, policy officials, and industries such as fisheries and aquaculture. To date, engagement with this broader group has been limited as communication pathways are less structured and often event-driven or relying on personal channels. This reflects the scientific team's primary focus on climate modeling rather than knowledge brokerage.

To strengthen stakeholder engagement in Year 2, this case study will concentrate efforts on AMAP as the primary stakeholder, building on the existing relationship with the Radioactivity Expert Group. Case Study 1 will also learn from Case Study 2's planned stakeholder workshop methodology and adapt relevant approaches for Arctic contexts. Participation in cross-case reflection sessions coordinated by Work Package 6 will facilitate knowledge transfer from more engagement-advanced case studies. Progress will be assessed in Year 2 project reporting (Deliverable 1.4).

3.1.5 Outputs, Timeline, and Synergies

An ideal scenario for this case study, outlined here as a result of the workshop, would be to have a completely automatized end-to-end process where an event alert triggers information flows that can be fragmented geographically and used by different stakeholders without the need to run R or Python scripts or have a technical or scientific understanding of data analysis, and yet, information is robust for decision-making at all levels. Potential outputs for this idealised scenario are as follows:

- An interactive visualization service, paired with explanatory materials, will present climate-driven changes in radionuclide distribution and exposure risks, making complex NorESM outputs accessible for AMAP expert groups.
- Crisis committees will continue to receive information about events and the potential disruptions to wildlife and arctic residents.
- Residents will be able to visualize realtime information about the quality of their water and other potentially contaminated resources in a simple interface.
- Local council officials will be able to access information similar to that of a crisis committee, but in near- to real-time and tailored to their specific locations and needs without having to run simulations or download large amounts of data.
- Industry and commercial sectors potentially disrupted by events receive necessary information to modify their activities according to the current and future levels of risk.

As of the production of this document, further engagement with stakeholders in this case study is planned to occur in 2026. This work will include prioritizing the requirements in collaboration with stakeholders.

This case study has potential synergies with Case Study 2 - RiOMar, as the main challenge of both is to convey large datasets to diverse stakeholders. Since the needs of both case studies focus around similar processes, it is conceivable that one of the results of this project will be a

data pipeline that is designed to subset data based on input end-user defined geographical polygons or bounding boxes on a platform that outputs different variables according to the profile of the user.

As other case studies advance, synergies may emerge. Interaction between scientific teams and external stakeholders could benefit from initiatives focused on capacity building and community-driven information flows, such as platforms like weAdapt, which offer alternatives to dashboards for sharing tailored information with Arctic residents.

3.1.6 Risks and Assumptions

A major risk is the difficulty of addressing FAIR data challenges while information already circulates in multiple contexts, reflecting common issues in climate adaptation and data governance. Tailoring outputs for diverse stakeholders is resource-intensive and hard to measure for impact; project driven dashboards often deteriorate over time due to lack of sustained funding for maintenance.

3.2 Case Study 2 - RiOMar Project (IFREMER): Observing and anticipating changes in French coastal areas influenced by rivers in the 21st century

Purpose:

1. Assess coastal water quality and marine ecosystem dynamics to anticipate and prepare for emerging climate impacts expected by the end of the century;
2. Identify risks and vulnerabilities using RiOMar model outputs as a foundation for developing scientifically based climate change adaptation (CCA) strategies.

3.2.1 Problem and Decision Context

This case study addresses the emerging risk of coastal ecosystem biodiversity loss due to water quality. The role of climate crisis disruptions on the diversity of ecosystems and the derived social impact of reduction of ecosystem services is still insufficiently understood. The knowledge gap in this topic is, to a large extent, a lack of available information that clearly describes the local impact of global climate change, and the interconnectedness of social local economies to biodiversity. While there is evidence that a connection between social systems and ecosystem biodiversity exists, the heterogeneity of such connection and the increasing role of climate crisis derived disruptions remains unclear. This case study intends to produce information that describes the disruptions of climate crisis events at a local level to empower stakeholders processes.

As the climate crisis progresses, we are losing opportunities to measure and evaluate the local conditions before disruptions. Similarly, the relationship between ecosystem diversity and socioeconomic processes at a local scale is rapidly changing worldwide. It is critical to observe and empower these transformations to the best extent of our understanding, and most importantly, to provide all the critical knowledge to local stakeholders so they can make informed decisions about their future, with knowledge about the possible impacts and uncertainties that exist.

Coastal ecosystems will greatly be affected by extreme events, possible increases of harmful algal blooms (impact on shellfish farming), and increase in environmental conditions that would be detrimental to biodiversity. Adaptation measures include changing upstream water management and farming regulations in the watershed, as well as spatial planning in coastal zones (where to plan shellfish farming, where to protect which kind of biodiversity). Decisions are to be made by local and national government agencies, as well as marine protected areas managers.

3.2.2 Actors

The main types of stakeholders involved in the project arepresented in Table A2 and summarized here:

- End users, who can be scientists, industry or public policy officials.
- Public institutions, who can act as data providers, facilitators.

Currently, the information is being used by research partners and institutions whose teams are experts and researchers. This project already engages with a core section of the stakeholders that have expressed specific needs that are sometimes beyond the scope of the project as it was planned.

3.3.3 Priority Knowledge Needs and Bottlenecks

This project's biggest challenge is that it is producing models that are very specialized (biochemistry), require technical knowledge to operate (large file sizes) and also operate in a very long window of time (30-100 years) and space (not easy to understand for local effects). One of the project's greatest challenges is to have stakeholders explicit which model outputs would actually be useful for them to make decisions, knowing that different stakeholders may have different needs.

3.3.4 Engagement and Methods

This case study has a strong collaborative relationship with scientific and technical partners and has engaged with diverse stakeholders including aquaculture professionals, water agencies, and coastal managers. However, the main challenge of this case study is translating these

interactions into structured inputs that can inform the FAIRification process. Without a systematic methodology for eliciting user needs (such as the design thinking approach applied in FAIR2Adapt), engagement activities have not yet produced the actionable requirements needed to extend project outputs to non-technical users.

Planned activities such as hackathons and Jupyter notebook demonstrations provide entry points, but a more structured approach to stakeholder engagement is needed to move from general interest to concrete co-design.

To address this in Year 2, IFREMER has recruited a dedicated team member to strengthen project involvement and coordinate outreach activities. Building on the stakeholder workshop planned in the project proposal, Case Study 2 will apply the design thinking methodology used in Case Studies 3, 4, 5 and 6 to structure engagement and elicit actionable user requirements. Participation in cross-case reflection sessions coordinated by Work Package 6 will support this methodological transfer. Progress will be assessed in Year 2 project reporting (Deliverable 1.4).

3.3.5 Outputs, Timeline, and Synergies

Ideally, this case study aims to provide information for a network of stakeholders in terms of current and future climate crisis driven scenarios and their impact on biodiversity. This case study in particular, aims to provide detailed information about specific outcomes for particular species, such as, the impact of increasing a few degrees or duration of heat wave, hypoxia event, or algae bloom, to know which species need protection and guide actions of adaptation. However, this case study has some challenges to achieve its potential. The most important ones are to include some of the key end user needs into the processes, and to extend the impactful interaction beyond the core scientific team of collaborators.

This challenge is also aligned with Case Study 1 - Arctic Radioisotopes and Case Study 3 - Hamburg City. There is a potential synergy between these cases and the other group of cases focusing on stakeholder engagement and capacity building, Case Study 5 - Connecting and visualising the climate change adaptation (CCA) knowledge landscape through the Climate Connectivity Hub (CCH) and weADAPT, Case Study 6 - Improving data availability for climate risk assessments under the EU taxonomy for sustainable activities.

3.3.6 Risks and Assumptions

This case study is also aligned with Case Study 1 - Arctic Radioisotopes in some of the risks and assumptions it faces to successfully implement end-to-end information based decision making for the climate crisis adaptation. The most important challenge observed in the workshops is that the information needed by some stakeholders who are potential end-users, and most importantly the expectations of these groups, seem to not be aligned completely with the core functionality of the models and the case study. This is a common hurdle in the

development of scientific knowledge for policy and implementation: timeframes. While the models are operating in large windows of time, decision makers need information that will enable them to make informed decisions in shorter timeframes. In an analogy often used in similar scenarios of climate change, having a forecast for the weather in 2 weeks does not inform whether it will rain tomorrow or the decisions I should make for the following days. Harmonizing these timeframes is one of the biggest challenges of this case study.

3.3. Case Study 3 - City of Hamburg: Climate-induced stressors and their effects on urban societies

Purpose:

1. Examine climate-induced stressors and their effects on urban societies
2. Integrate socio-economic data on vulnerability and exposure of urban population with physical climate hazard data to explore current and future climate risks of the city of Hamburg and to inform sustainable CCA strategies.

3.3.1 Problem and Decision Context

This case study intends to understand urban climate vulnerability from an integrated perspective. To do so, they have developed novel and unique methods to combine multiple layers of hazard, vulnerability and exposure to produce a diverse range of risk indexes for the city of Hamburg. Their methods combine population microdata layers with unique measures such as specific building heat indexes. This results in a unique layer that produces actionable insights for public policy, specially for the mitigation and adaptation to extreme weather events such as heatwaves. This case study is also strongly articulated with a diverse range of stakeholders in the city of Hamburg, and the case study personnel is familiar with the procedures in the city and the decision-makers, having in many cases a personal connection and knowledge about each individual in a particular role, which is one of the biggest challenges in knowledge translation into informed decision-making.

Understanding the heterogeneity of urban risk is crucial to mitigate and adapt to the climate crisis. For example, understanding the specific heat index of different urban infrastructure can enable the repurposing of buildings such as churches during heatwaves. Most importantly, having the ability to overlay layers of these buildings with microdata population layers enables the prioritization of interventions across the city. These are explicit actions that can be triggered by the methods developed in this case study and that have the potential to save lives.

As extreme weather events will become more frequent and more intense, it is crucial to understand how the heterogeneity of heat indexes operate in a city, and to what degree having a nuanced understanding of high resolution vulnerable populations can help adaptation efforts. In a way, this case study is posed to help us understand the degree to which information based decision making can compensate for infrastructure driven climate risk. In other words, how

much we need to change our cities to new realities will be determined by the degree to which our understanding of risk can help us use existing infrastructure in novel ways to mitigate emerging risks. This project is unique in its applicability window, offering unique insights both for short-term implementation and knowledge for long-term adaptation as the impact of information-based decision making is understood.

3.3.2 Actors

The stakeholder landscape for this case study includes a wide range of actors involved in providing, managing, and using adaptation information. Data providers such as the City Agency for Geoinformation (LGV) and governmental scientific bodies supply accurate datasets and expert analysis, feeding into the city's administrative and budgeting workflows. End users include the Senate, which prioritises budgets and evaluates impacts, and the Climate Adaptation Coordination Unit, which requires reliable, comparable, and easily shareable information to identify potential intervention sites. Researchers—professors, students, and modelling teams—develop risk assessments that depend on well-structured metadata and consistent inputs. Civil society, including citizens and community groups, needs adaptation information that is accessible and digestible. Together, these actors illustrate a complex ecosystem in which scientific accuracy, usability, and institutional workflows must align for adaptation decisions to be effective.

3.3.3 Priority Knowledge Needs and Bottlenecks

This case study has developed a sophisticated model and generated actionable insights, with some of its outputs already informing interventions. It operates within a well-established ecosystem of trust, where stakeholders share a common understanding of adaptation needs and processes. Yet despite this strong foundation, translating technical advances and long-standing relationships into consistent, end-to-end decision support remains the central challenge. Political dynamics and institutional constraints limit the extent to which the available models can be fully integrated into policy. The biggest challenge of this case study is moving from successful pilots and isolated impacts toward a coherent, FAIR-enabled framework that reliably connects knowledge to urban adaptation decisions.

3.3.4 Engagement and Methods

Engagement in this case study is centred on continuous collaboration with the municipal institutions responsible for climate adaptation, spatial planning, and risk management in Hamburg. Rather than formal consultations or periodic workshops, the process is built around working sessions where maps, datasets, and analytical methods are reviewed directly with the city's technical officers. These exchanges allow the team and municipal partners to jointly test

the usability of outputs, identify where data sensitivity limits application, and clarify how scientific information aligns with the city's decision-making cycles.

Interactions have taken the form of focused technical discussions on specific outputs, such as the pluvial flood-risk map, and methodological reviews where assumptions, data sources, and modelling decisions are openly examined. This approach ensures that each iteration of the work is anchored in how the information will be used in practice. Engagement also extends beyond Hamburg, with exploratory contacts initiated with the City of Graz to assess the potential transferability of methods and data structures developed in this case.

Feedback from these exchanges is directly shaping the next steps. The team is preparing a peer reviewed publication of the pluvial flood risk map, together with a methods paper that documents the main decisions and includes a reproducible example for transparency and replication. In parallel, a FAIR Implementation Profile and a research-object container (RO-Crate) are being assembled to link data, code, and metadata in a coherent structure. A revised manuscript incorporating stakeholder feedback is also being developed to reflect the project's evolving understanding of how the FAIR principles can support urban adaptation.

One of the main challenges identified through these interactions is translating scientific products into formats that are operationally relevant for planners and decision-makers. To address this, the team is exploring simplified versions of the maps, as well as comparisons between new heat risk layers and the city's existing "cool places" maps to highlight gaps in cooling infrastructure. In practice, this collaboration has already led to informal use of the pluvial flood risk map to inform the allocation of flood risk reduction resources, providing an early indication of how these tools can support local action.

Engagement will continue through the next project phase, maintaining direct exchanges with city departments while expanding collaboration to potential transfer sites and related case studies within the consortium.

3.3.5 Outputs, Timeline, and Synergies

The main outputs of this case study focus on spatial products and analytical materials that support climate adaptation planning in Hamburg. The most immediate result is a peer reviewed pluvial flood risk map that combines multiple data layers, including climate hazard, exposure, and vulnerability, at a resolution that can be directly used by the city. This map is both a scientific contribution and a decision support tool, providing an updated understanding of flood risk under changing climatic conditions. It will be accompanied by a detailed methods paper that explains the analytical approach, the assumptions made, and how the information can be interpreted and applied by practitioners. Both outputs are expected during 2026.

Alongside these scientific products, the team is preparing a revised manuscript that integrates the feedback and observations of city officials and technical officers. The purpose is to align the

description of results with the practical needs of those who will use them, ensuring that the methods and findings are communicated in a way that supports local decision-making.

The main users of these outputs are the city departments involved in spatial planning, environmental management, and infrastructure investment, particularly the Senate of Hamburg, the LGV (City Agency for Geoinformation), and the Ministry for Environment, Climate, Energy and Agriculture. Other actors, such as consultants, and local research groups, are also expected to draw on these products for risk evaluation and urban planning studies.
These outputs directly support decision processes related to the allocation of adaptation budgets, the identification of areas requiring cooling infrastructure, and the prioritisation of flood risk reduction investments. The work also contributes to ongoing policy discussions about how to reconcile adaptation measures with other urban development goals such as densification and infrastructure renewal.

This case study connects closely with others in the project, particularly with Case Study 6 (Adelphi), which focuses on translating scientific data into policy and financial language. These collaborations aim to understand how scientific information can be made actionable across different governance and policy contexts.

3.3.6 Risks and Assumptions

One of the main risks for this case study is the difficulty of moving from scientific results to institutionalised use. Much of the collaboration in Hamburg relies on personal relationships and informal exchanges between researchers and public officials. While this has enabled a high degree of trust and flexibility, it also makes the process vulnerable to staff changes or shifting institutional priorities. To mitigate this, it is important to document processes and decisions clearly, ensuring that the methods, data, and results can remain accessible and useful beyond individual partnerships.

A second risk relates to the fragmentation between scientific production and decision timelines. Municipal planning and budget processes often move faster or in different cycles than scientific research, meaning that outputs may arrive too late to inform key decisions or too early to be absorbed. For example, the time taken to publish the article, which is one of the main outputs of this case study, can be a critical time to pilot and implement measures derived from the knowledge produced in the case study. To mitigate this risk it would be important to maintain open communication with city offices and to share intermediate products, even when they are still under development, so that feedback and partial use can occur in real time.

A third risk concerns data sensitivity and privacy, particularly when working with microdata at the building or neighbourhood level. The very detail that makes these analyses valuable for adaptation planning also raises questions about data protection and public communication. To

address this, the case study is testing simplified and aggregated map versions that preserve the analytical insight while reducing the risk of exposing sensitive information.

Together, these risks underline a broader challenge shared by many adaptation efforts: ensuring that robust scientific knowledge is not only produced but embedded in the institutional systems that need it.

3.4. Case Study 4 (FCID): Designing a FAIR National Adaptation Hub

Purpose:

1. Concentrate and make FAIR, the wealth of resources typically considered relevant for national-to-local adaptation planning (e.g., scientific and grey literature, datasets, policy documentation, knowledge platforms).
2. Develop a FAIR-by-design platform that contains these FAIRified resources and make it available to adaptation relevant actors in Portugal.
3. Apply, in practice, the FAIR2Adapt approach to design Portugal's first National Adaptation Hub, and test its relevance for adaptation policy decision-making process at multiple scales.

3.4.1 Problem and Decision Context

Across Europe, countries and regions fare under growing pressure to mitigate climate impacts and build additional resilience in response to observed and projected climate changes. In recent years, the EU has strengthened its commitment to climate action through the EU Climate Law (EU, 2021/1119), as a part of the European Green Deal from 2019, which legally binds Member States to achieve climate neutrality by 2050. The 2021 EU Strategy on Adaptation further calls for a smarter and more systemic approach to building resilience, urging national and regional governments to revise their adaptation policies accordingly.

Although numerous adaptation portals and platforms exist to support policy and practice—such as Climate-ADAPT, MIP4Adapt, and various global data services—these resources remain underutilised (European Commission, 2025). Reasons for this could be related to the profusion of platforms and knowledge hubs already existing, which causes confusion amongst stakeholders about good practice and what is relevant, salient and credible information. Portugal faces similar challenges. Since the adoption of its first National Strategy for Adaptation in 2010 (Resolução do Conselho de Ministros nº 53/2020), the country has generated a substantial body of scientific outputs, policy documents, datasets, and grey literature, yet many of these resources remain difficult to find or access for the diverse community involved in climate adaptation.

The proposed Portuguese National Adaptation Hub aims to address this gap by focusing initially on heat-related hazards—a growing hazard with serious impacts on population health and

urban ecosystems. By consolidating and FAIRifying key datasets, documents, and knowledge on heat-related risks (Table A4), the Hub seeks to make adaptation resources more discoverable, interoperable, and usable. Its purpose is to streamline the flow of information across researchers, practitioners, and decision-makers, supporting more effective planning and implementation of adaptation measures throughout the country.

3.4.2 Actors

The Portuguese National Adaptation Hub is designed to serve a wide and diverse community of actors who produce, translate, and apply climate adaptation knowledge (Table A5). These include data producers from research institutions and private companies, who generate the scientific evidence and datasets underpinning adaptation decisions but often remain disconnected from the needs of end-users. Knowledge producers such as consultants, NGOs, and governmental agencies interpret and contextualise this information, though they frequently struggle with fragmented regulatory frameworks, outdated or inaccessible data, and limited machine-readable formats. Policy owners at the national level depend on this knowledge to update sectoral strategies, while networks like Adapt.local link data providers, municipalities, and practitioners to help close the gap between scientific knowledge and local-level action.

At the same time, a broad group of end-users—including municipal executives, adaptation planners, businesses, auditors, media actors, and local communities—require reliable, up-to-date, and easy-to-understand information to support everyday decisions. Many of these users face significant challenges: complex administrative processes, data that is not packaged for practical use, limited access to the latest policy documents, or difficulty interpreting technical materials. By acting as a knowledge broker across all of these groups, the Hub aims to streamline the flow of information, improve FAIR data management, and make adaptation-relevant knowledge more discoverable and usable. Ultimately, it seeks to bring coherence to a fragmented landscape, connecting national-level datasets and scientific outputs with the needs and realities of practitioners, decision-makers, and citizens.

3.4.3 Priority Knowledge Needs and Bottlenecks

The Portuguese National Adaptation Hub aims to provide end-users with meaningful, updated, and accessible knowledge for climate adaptation. To achieve this, the FCID team will identify, extract, and organise a broad set of climate and non-climate resources (Table A4). These materials will support decision-making processes across Portugal’s adaptation community. For example, an NGO working on a LIFE project may need recent information on water use and management in the Alentejo region to guide livestock-sector adaptation efforts, illustrating the importance of centralising and updating relevant knowledge.

Once these resources are identified, they will be FAIRified into interoperable and reusable FAIR Digital Objects (FDOs) using frameworks developed in WP2–WP4. This process will help align vocabularies, correct mismatches between data producers and practitioners, and enable seamless connections between datasets, knowledge objects, and executable digital tools. The Hub itself will be designed with a strong focus on user experience (Figure A1), ensuring that knowledge is both interpretable and ready for processing by diverse actors—from municipal practitioners to journalists. FAIR-enabling services such as ORKG reborn and RO-Hub will support this functionality, making it possible for users to filter policy documents for media reporting or replicate analyses described in scientific articles for local adaptation planning.

3.4.4. Engagement and Methods

Adaptation knowledge needs were assessed, at a first stage, by engaging with the Case Study 4 owners from the FAIR2Adapt consortia, namely the FCID partners. Case study owners gathered in internal meetings and in a workshop organized by WP6 and WP5 partners (please see methodology in section 2 of this document).

All relevant resources for the National Hub are being listed and retrieved by the CS4 team members. To identify and extract all relevant peer-reviewed articles, a systematic search was performed across Scopus and ISI Web of Science for articles between January 2010 and May 2025. The results were analysed by geographical scope, adaptation sectorial focus, knowledge sector, climate hazard and CCA needs. Alignment of the current scientific outputs with FAIR principles was also analysed.

3.4.5 Outputs, Timeline, and Synergies

Policy documents and other grey literature is being listed and extracted by the CS4 team based on expert knowledge and personal libraries (in instances when documents are no longer available online due to broken links). Knowledge platforms and portals will be identified by the CS4 team in conjugation with the data retrieved (and/or used) by the scientific articles. A complete list of all resources for inclusion in the Hub is expected by March 2026 (Figure A2). The next step will be the presentation of the Hub and the resource list to a group of stakeholders, possibly a part of the Adapt.local community, during a workshop. The main objective of the workshop will be to pitch the Portuguese National Adaptation Hub to the community and collect feedback on any missing data and knowledge gaps in the Hub (Figure A2).

Several similarities and synergies were found between the Portuguese National Adaptation Hub and other Case Studies, namely CS5 (Connecting and visualising the CCA knowledge landscape through the CCT, CCH and weADAPT) and CS 6 (Improving data availability for climate risk assessments under the EU taxonomy for sustainable activities). The vocabulary for the adaptation sector based on the weAdapt taxonomy (CS5) is currently under improvement (WP5). The Hub can use the vocabulary when tagging resources. Both CS4 and CS6 will require the translation of grey literature (such as municipal adaptation strategies) into

machine-readable information. The collaboration with other workpackages, such as WP4 which uses ORKG Reborn and WP3 which is creating the FAIRification process, could help accomplish this. It is probable that CS6 will develop its work on this theme before CS4, and therefore CS4 will follow up on CS6 updates and ways forward.

3.4.6 Risks and Assumptions

A number of risks and assumptions have been identified by the CS4 team that may influence the progress and outcome of this case study.

First, as the Hub depends on externally hosted materials, broken or outdated resource links pose a risk to its integrity and usability. There is a dependency on third-party sources remaining active and accessible.To mitigate this, archiving strategies will be explored to ensure access.

As a second risk, some resources may present insufficient metadata, which may limit the creation of FDO's and overall potential for FAIR compliance. To mitigate this, an ongoing curation and metadata enhancement will help fill any gaps to maintain data quality and ensure FAIR standards are maintained across the Hub.

As a third risk, there are limitations in FAIRifying certain resource types, particularly grey literature such as policy documents, reports and qualitative datasets, as they often lack structured data formats or standard metadata fields, which can constrain their alignment with FAIR principles. The assumption is that partial FAIRification, combined with tagging and enhanced metadata will improve their findability and reusability.

As a fourth risk, limited engagement and external user uptake may constrain the Hub's impact and reach. The assumption is that targeted communication will help stimulate engagement. Additionally, by FAIRifying these resources, it is expected that they will become more findable and accessible across platforms beyond the Hub itself, therefore contributing to the broader adaptation community.

Finally, this case study focused solely on heat-related hazards and associated impacts, even though several other hazards are also being addressed in national and regional adaptation strategies in Portugal (e.g. forest fires, floods, sea level rise). By focusing on heat-related hazards at this stage, the case study can benefit from methodological depth without putting aside the future inclusion of other climate hazards in the Hub, through the application of the methodological framework designed in FAIR2Adapt.

3.5. Case Study 5 - Connecting and visualising the CCA knowledge landscape through the Climate Connectivity Taxonomy, Hub and weADAPT.

Purpose:

1. Improve the Climate Connectivity Hub (CCH) and Taxonomy (CCT) to support knowledge managers, tool developers and platform owners (e.g. weADAPT) to facilitate access and sharing of climate-resilient adaptation strategies that inform local, national, regional, and global policy processes for communities, planners, practitioners, students, researchers, and policymakers

2. Break down silos and avoid redundancy, replication, and wasted resources while promoting collaboration, dialogue, and learning by harnessing the wealth of existing knowledge across various portals, platforms, and publications, and providing guidance on how to connect this knowledge together.

3. Integrate the CCT, Hub and weADAPT in the EOSC.

3.5.1 Problem and Decision Context

As outlined in the introduction, one of the central challenges for adaptation today lies in the final stage of the end-to-end decision-making process, where information must be transformed into decisions and concrete action. In many cases, relevant and timely information already exists, but it remains siloed, inaccessible, or presented in ways that are not easy to interpret or apply by decision-makers. The gap is rarely due to a lack of data: it lies in the way knowledge moves (or fails to move) across institutions, sectors, and languages of practice.

This case study addresses that gap by strengthening the role of weADAPT and the Hub as spaces that connect information, people, and processes. These platforms have positioned themselves as facilitators in a complex ecosystem of stakeholders that includes regulatory bodies, multilateral organisations, governments, NGOs, research institutions, and local communities. Together, they provide a participatory framework that recognises that adaptation depends as much on inclusion and communication as it does on data.

Through courses, partnerships, and capacity-building activities, this case study focuses on connecting specialised actors with the audiences who most need their insights. It turns scattered knowledge into shared understanding, supporting the idea that adaptation is not only a technical process but also a social one, a collective effort to make knowledge actionable where it matters most.

3.5.2 Actors

The diversity of actors represented in this case study highlights that adaptation knowledge does not move in a straight line from science to policy (Table A6). It circulates across an ecosystem that includes producers, brokers, and users, each with their own timeframes, languages, and incentives. The Hub, CCT and weADAPT operate at the intersection of these worlds, creating the links that make information usable and traceable. Policy makers and funders need concise, defensible evidence they can act on quickly, while NGOs, community organisations, and educators need adaptable materials they can translate and reuse in their own contexts. Researchers and modellers need clearer channels to communicate their findings beyond academia, and communities need to see their lived experience recognised as legitimate knowledge rather than anecdote.

Across all actors, a common pattern emerges: every group is simultaneously a producer and user of knowledge, and their ability to act depends on whether information flows are reciprocal, trusted, and timely. This case study stands out for its long-standing experience in co-creation — working directly with policy makers, researchers, NGOs, and communities to turn knowledge into action. Through the Hub, CCT and weADAPT, it demonstrates how engagement with the full ecosystem of adaptation actors can make policy implementation more inclusive and grounded in real needs. At the same time, this experience exposes one of the biggest challenges of today's climate work — and one shared by several other case studies in FAIR2Adapt: how to sustain these collaborative processes so that the bridge between knowledge and decision-making remains active long after individual projects end.

3.5.3 Priority Knowledge Needs and Bottlenecks

This case study faced one of the most complex challenges of all: how to translate the richness of qualitative information such as community knowledge, training materials, conversations, and lived experiences into FAIR artifacts that can be shared, traced, and reused. This difficulty cuts to the core of climate adaptation, where scientific and policy information often struggles to match the pace, language, and trust-building needed for genuine co-creation. The CCH and weADAPT operate precisely at this intersection, working to align technical standards with the human processes of learning and collaboration.

The main knowledge need emerging from this case is how to make participatory and qualitative processes FAIR without losing their narrative depth or context. Stakeholders working with communities require information that reflects local realities and values, not just as inputs for diagnosis, but as active components of the solutions. However, scientific and policy frameworks often move faster than the cultural processes through which communities build resilience. When this gap widens, trust erodes and adaptation measures risk becoming symbolic rather than transformative.

To address this, the case study is prioritizing the FAIRification of the Hub and CCT as a shared vocabulary for climate adaptation. While the challenge of producing FAIR qualitative artifacts was discussed in this case study, it remains of the pressing challenges and a future avenue of development for future prioritization.

In practical terms, the bottleneck is not the absence of information but the difficulty of structuring diverse knowledge in a way that retains meaning across scales and contexts. This case demonstrates that building systems that honour cultural sensitivity, participation, and co-creation is both the greatest challenge and the greatest opportunity for the next generation of adaptation knowledge infrastructures.

3.5.4 Engagement and Methods

Engagement in this case study is defined by the decision to focus first on producing a common vocabulary for adaptation, rather than immediately FAIRifying the full set of weADAPT resources. The FAIRification of weADAPT itself, including community stories, practice notes, training materials, and other qualitative content, will take place at a later stage in the project. At this stage the main effort has been to further develop v1 of the CCT that can act as a shared language across the consortium.

This CCT is being prepared as a FAIR vocabulary so that it can serve as an underlying connection between the different FAIR artifacts produced in the other case studies. In practice, this means that instead of starting with outreach to a broad set of external stakeholders, most of the engagement so far has been with technical partners inside the consortium, especially the teams working on data structures and interoperability. The aim is to ensure that the terms and relationships defined in the CCT can be used consistently, and can describe both quantitative outputs (for example, hazard types and adaptation measures) and qualitative outputs (for example, policy guidance or community practice).

These interactions have taken the form of working sessions to review terms, align definitions, visualization tools, FAIR vocabulary standards and test how concepts from real adaptation practice can be represented in a structured way. Input from previous community engagement in other EC projects including training and engagement with Real World Labs in the Directed project is feeding into this process so that the controlled vocabulary still reflects real-world language and not only internal technical choices. Furthermore, the vocabulary is structuring these platforms so that end-users seamlessly interact with FAIR artifacts and structures without directly needing to address technical specifications, but dealing with information that is easier to find and access.

As the taxonomy stabilises, engagement will shift again toward the external users of the platform. The plan is to validate that the vocabulary helps practitioners find relevant material faster, helps policy makers access actionable examples, and helps communities surface and share their own work in ways that can be seen and reused. In other words, work in this first

phase is about building the connective tissue that will allow future FAIRification of the platform to actually benefit its users.

3.5.5 Outputs, Timeline, and Synergies

The main output of this case study for the first half of 2026 will be the development of a FAIR vocabulary for CCA. This vocabulary will form the foundation for future FAIRification of the weADAPT platform, providing a consistent language that connects the diverse datasets, models, and narratives emerging across the FAIR2Adapt consortium. It will allow the knowledge generated by different partners to be more easily discovered, linked, and reused, serving as the underlying structure that ties together the project's varied outputs.

The FAIR vocabulary will be developed in close coordination with other technical work packages, particularly WP2 and WP3, to ensure semantic and metadata consistency across systems. Once operational, it will support both the backend integration of FAIR artifacts and the future user-facing functionalities of weADAPT, enabling better search, navigation, and knowledge interlinking across adaptation topics. The main users of this output will initially be internal (researchers, data curators, and technical partners) but the benefits will gradually extend to external audiences such as practitioners, policy makers, and educators who rely on weADAPT to find and share relevant adaptation knowledge.

In parallel, the case study will explore synergies with other case studies in the consortium. For example, weADAPT can serve as a microsite or dissemination space for Case Studies 1, 2, and 3, which each produce highly specialized datasets and models that would benefit from wider community engagement and circulation. By hosting or linking these outputs, weADAPT can act as the bridge that connects technical knowledge with stakeholder-driven processes, ensuring that the scientific work of the consortium reaches practitioners, decision makers, and local communities in accessible and meaningful ways.

In the broader context of FAIR2Adapt, this case contributes to the goal of creating a connected, interoperable ecosystem for adaptation knowledge. While the early work focuses on the technical layer such as defining terms, structures, and linkages, its ultimate purpose is to enhance participation, learning, and collective decision-making across all other cases. During 2026, the FAIR vocabulary will provide the common backbone that enables each case study to contribute to a shared knowledge landscape, turning fragmented data and experiences into a coherent, searchable, and FAIR ecosystem.

3.5.6 Risks and Assumptions

One of the main risks for this case study is that much of the current work is technical and backend-oriented, which can temporarily distance it from the wider user community that normally interacts with weADAPT. The process of developing a FAIR vocabulary requires

sustained coordination with internal partners and may not immediately produce visible results for practitioners or policy makers. To mitigate this, it would be important to maintain different parallel lines of work so that the development of the CCT does not overshadow the very important contribution that this case study can provide to the consortium, which is enabling the expansion of the stakeholder engagement through new dissemination channels for the specialized technical information.

A second risk is that the FAIR vocabulary may not fully capture the diversity of concepts used across disciplines and languages in climate adaptation. Since adaptation is a context-specific and culturally sensitive process, there is a risk of creating a rigid or overly technical structure that limits inclusion. To address this, the vocabulary aims to remain open and iterative, allowing for feedback, modification and the gradual inclusion of new terms as they emerge from other case studies and stakeholder engagements.

3.6. Case Study 6 - FAIRification of a Climate Adaptation Strategy: Improving the accessibility and usability of adaptation information for policymakers

Purpose:

1. **Provide accessible information on climate change risks and adaptation** based on the cities adaptation strategy and risk assessment to city departments, citizens, businesses and other stakeholders in Bremen

2. **Raising awareness of the urgency of climate adaptation and the effects of climate change on Bremen/Bremerhaven**: The inclusion of storytelling elements is intended to help make the need for analysis and action understandable, thereby increasing acceptance among the population and decision-makers.

Remark:

Case Study 6 moved from the objective to consult small and medium businesses on how to prepare climate risk assessments aligned with the FAIR principles for sustainability reporting towards the current, here described, scope, supporting the city and federal state Bremen in FAIRifying climate risk and adaptation information for different stakeholder groups. This change was due to amendments in EU regulations (CSRD reporting) and thus the difficulty in finding companies that despite the changes still wanted to conduct CRAs for their sustainability reporting. By focusing on communal stakeholders and FAIR adaptation strategies as we do now, we make sure to cover several parts of the adaptation cycle within this case study.

3.6.1 Problem and Decision Context

The partner in this case study is the Free Hanseatic City of Bremen, represented by its employees in Department 43, who are responsible for climate adaptation in the federal state and city.

Bremen already has a number of key documents relating to climate adaptation. These include the Bremen-Bremerhaven Heat Action Plan 2024, the Agriculture Development Concept 2035, the Biodiversity Strategy and Insect Protection Programme 2024, the Climate Adaptation Strategy for Bremen and Bremerhaven 2025, an urban climate analysis and the DWD climate report for the state of Bremen. The city's website also provides small-scale data on observed and expected climate changes and on monitoring of various climate indicators. The German Weather Service, the Geological Service of Bremen, the Statistical Office of Bremen and the Social Affairs Department were used as key sources of information and data for the development of the climate adaptation strategy. The main target groups of the current climate strategy are the public administration of the city of Bremen and the general public.

Building on Bremen's strong foundation of climate policy documents and existing data infrastructures, the project develops user-centred methods to combine multiple layers of hazard, exposure, and sensitivity into practical risk insights for different stakeholder groups.

The city of Bremen aims to improve the implementation of its climate adaptation strategy. Those responsible see particular potential in improving awareness, simplifying accessibility and making content more accessible to all citizens as well as showcasing adaptation measures and efforts.

Understanding the heterogeneity of urban risk across Bremen's neighbourhoods is central to prioritising adaptation and engaging citizens, private sector and local advocates.

During an initial workshop in the FAIR2Adapt project, a vision, a stakeholder analysis and initial steps for future implementation were developed together with the participants from the State Centre for Climate Adaptation.

The goal is thus to prepare data and information in a target group-oriented manner via a stakeholder-oriented dashboard, that allows district advisory boards, youth representatives, city administration, politicians, and businesses to quickly explore tailored views of their city quarter in terms of climate risks and adaptation efforts and trigger concrete actions with simple narrative "explainers" that improve awareness and accessibility for all citizens.

3.6.2 Actors

During a workshop with representatives from the City of Bremen's climate adaptation department (Referat 43), several key actors involved in local adaptation processes were identified and discussed (Table A7, Figure A3). Referat 43 plays a central coordinating role, responsible for strategic steering, integrating diverse datasets, and balancing the often

competing needs and priorities of multiple municipal departments. Their goal is to enable effective, cross-sector adaptation with broad stakeholder participation, yet they face significant challenges related to bureaucratic constraints, data dependencies, and navigating differing institutional interests—for example, reconciling adaptation plans with the requirements of the heritage conservation authority. Local advocates, represented by the Beirat für Stadtteil, serve as intermediaries between municipal structures and neighbourhood-level communities. They aim to relay tailored, actionable information to residents and to ensure community voices are included in decision-making. Their effectiveness, however, depends heavily on the level of engagement and capacity of the individuals involved.

Additional actors include data stewards such as GeoInfo and SWB, who develop, manage, and share essential datasets. While they aspire to provide reliable and integrated data flows, their work is hampered by siloed systems and limited interaction with Referat 43, including restrictions around metadata sharing and access to certain datasets. Engaged citizens also play an important dual role as both recipients and drivers of climate adaptation. They seek clear, accessible information that enables participation in local processes, yet they often encounter barriers due to the complexity of adaptation data and a perceived lack of transparency. Across all these groups, the workshop highlighted a need for more open data flows, stronger cross-departmental collaboration, and communication formats that make adaptation information easier for non-experts to understand and use.

3.6.3 Priority Knowledge Needs and Bottlenecks

1: Lack of integration and coordination - Municipal utilities, geoinformation services, and other data controllers often work independently. As a result, there is a lack of central integration and closely coordinated processes and information. Different responsibilities and priorities make cooperation difficult, e.g. for data requests/data and information management.

2: Unclear framework conditions - There are uncertainties regarding data use (including rights) and transfer in addition to **complex coordination processes for data requests/data and information management**. Multiple data controllers with different priorities; data protection rules and licenses.

3: Lack of interfaces, information sharing, and involvement - District councils, citizens, and government agencies currently have only limited and user-unfriendly access to information and data; often, there is no district-specific information available. This can make it difficult to support adaptation measures in private and public spaces. Fragmented communication channels; accessibility requirements (language, readability, disability); trust gaps; constrained communications capacity.

3.6.4 Engagement and Methods

- Workshops on decision need basis with Referat 43 and local advocates together with actors from health department (1 interview, 2 workshops)

- Regular exchanges for dashboard development to ensure a participatory process (joint discussion on contents, user experience, visualization);
- The first stakeholder mapping exercise took place with tarot mapping (design thinking tool, Figure A3 to A6)

Engagement in this case study is centred on continuous collaboration with the municipal institutions responsible for climate adaptation, spatial planning, and risk management in Bremen. There will be consultations together with dedicated workshops, to identify which datasets are most relevant, what information is needed most in city quarters and how information should be presented best. Translation of policy documents and respective relevant data to actionable information through a dashboard can be challenging. We need to make information relevant to city stakeholders and therefore came up with “two” versions of a dashboard (professional and non-professional), meaning the simplification of information, adaptation measures (Figure A6). By asking questions about the stakeholder needs we keep them involved and assist by filling gaps of knowledge and capacities within this case study. With clear structured protocols of all meetings, that are circled around for all participants, we guarantee all thoughts and inputs from the stakeholders are documented and contribute to the further progress. Feedback from these exchanges is directly shaping the next steps. When a first draft of a dashboard is developed, we plan to test it intensively with the stakeholders from Bremen to double check whether this tool is meeting their requirements.

Possible integration of other stakeholders next to our contact partners at Referat 43 can be more challenging. We trust in Referat 43 to reach out to persons that are relevant to talk to. By using their network and contacts we can reach the citizens or other departments when necessary. Engagement will continue through the next project phase, maintaining direct exchanges with Bremen while expanding collaboration to related case studies within the consortium.

3.6.5 Outputs, Timeline, and Synergies

The case study’s aim is to translate existing knowledge on climate risks and adaptation measures provided by policy documents to citizens and other municipal, civil society and private sector stakeholders. It is intended to provide the information accessible via the development of a user-friendly dashboard that allows users to select districts and clearly displays various maps and relevant data based on the climate risk and adaptation strategy of the city of Bremen, as well as providing information on adaptation, in order to make the large amounts of data (which are already available but only accessible to a limited extent due to different reasons in terms of data ownership, licensing and data protection and usage concerns) usable for different stakeholders in a targeted manner through a FAIR process, so that also in the future, methods, data and results are accessible and regularly updated. The dashboard should focus in particular on neighbourhoods and involve local stakeholders as multipliers in order to convey information clearly. Important social infrastructure will be given special consideration. The results and products, such as visualised risk analyses or PDF reports, should be easily accessible and

usable via the dashboard (e.g. Dashboard City of Rotterdam). FAIR2Adapt aims to assist Bremen in processing and presenting existing data and findings on climate adaptation in a new format so that additional target groups can be reached. This can be achieved through an online dashboard.

In addition to an easy-to-understand basic version with clear graphics and instructions for action of a dashboard, a "professional version" with extended data options, filters and download functions is also offered. Storyboards illustrate complex relationships such as sponge city principles.

A possible structure for the dashboard includes an overview page introducing its objectives and offering both a basic and a professional version, along with language settings in German and English. Core content would combine maps, graphics, and data sourced from Bremen's main providers—including downscaled climate projections from the German Weather Service, green space and bioclimate information from the State Statistical Office and Geological Service, social facilities and heat hotspots from the Social Affairs Department, and sponge-city elements such as water retention areas. Users could select districts or freely defined areas through an interactive map, apply filters (e.g., climate risk class, sealed surface, green space distribution), view key figures highlighting hotspots and priority areas, and access tailored recommendations for action along with responsible authorities and contact points. The planned timeline foresees consultations on technical feasibility by October 2025, a joint coordination meeting in November, a stocktaking report by December, and a rough dashboard concept by the end of 2025. In 2026, a stakeholder workshop or survey will be held in Bremen to refine user requirements, followed by a first mockup in March and a presentation to local quarter managers in April or May. Many of the challenges mirror those identified in the Hamburg case study and the National Adaptation Hub, particularly the difficulty of transferring information to citizens, local actors, and the private sector; the dashboard is therefore intended to reinforce the translation of policy into locally meaningful action.

3.6.6 Risks and Assumptions

The integration of the dashboard into the city's IT environment assumes structured, transparent collaboration between Bremen's IT teams and FAIR2Adapt technical staff; without this, misalignment on architecture, security, or deployment could cause delays or rework. Successful use of datasets owned by external providers relies on early alignment workshops with data holders such as SWB, GeoInfo, the Statistical Office, and the Social Affairs Department to establish permissions and formats, while restrictive sharing policies, heterogeneous standards, or licensing constraints could slow or limit data flows. On the implementation side, a dedicated workshop is expected to define realistic dashboard options that match available technical capacity, though insufficient development resources or competing priorities could still lead to delays or reduced functionality. The long-term sustainability of the dashboard depends on a clear maintenance plan with defined ownership and budgets; without this, pipelines could break, security could weaken, and user trust could erode. Uptake also requires active contributions

from local stakeholders, meaning that limited capacity building or unclear responsibilities could hinder effective use. Data security and licensing issues must be settled early with data providers and authorities, as heightened concerns could restrict access to essential datasets or prolong review processes. Finally, the inclusion of potentially sensitive data requires careful distinction between internal and public-facing views, with appropriate privacy-preserving aggregation, since releasing granular or misinterpreted information could stigmatise communities, trigger backlash, or undermine trust.

4. Comparative Insights and Expected Impact

The following visualization (Figure 1) summarises the full end-to-end flow of data, stakeholders, and processes identified across all six case studies.

The six case studies in FAIR2Adapt together illustrate the extraordinary diversity and the shared challenges of building data-informed adaptation systems in practice. They range from complex Arctic and marine models that simulate long-term climate impacts, to urban-scale tools for immediate public action, to participatory networks connecting local communities and policy-makers through digital platforms. Despite this diversity of contexts and scales, all case studies converge around a common goal: ensuring that climate knowledge becomes usable, trusted, and timely enough to inform real decisions.

A comparative reading of the case studies reveals two broad categories of adaptation challenge. The first revolves around scientific translation: how to convey vast, technical, and model-driven datasets in a way that policy-makers, practitioners, and communities can interpret and act upon. This is the core tension visible in the Arctic, RiOMar, and Hamburg cases. Each has developed sophisticated modelling or mapping tools that represent climate impacts with high spatial and temporal precision. Yet, as the workshops revealed, producing accurate information does not necessarily translate into actionable decisions. Instead, effective adaptation depends on contextualising information so that it aligns with the language, timescales, and incentives of those expected to act.

The second category concerns stakeholder articulation: how to coordinate the multiplicity of actors that must work together for adaptation to happen. This is evident in the Portuguese Adaptation Hub, the CCH and weADAPT, and the Climate Risk Assessment cases. These focus less on producing new data and more on enabling communication across fragmented networks. Their challenge is to align vocabularies, workflows, and data formats so that different types of knowledge - scientific, technical, institutional, and community-based - can coexist and reinforce each other.

Across both categories, a recurrent insight emerged: adaptation is not only about data integration but about institutional and cultural integration. In every case, the point where information meets decision-making is also the point where social trust, organizational routines, financial and time constraints collide. This explains why many of the bottlenecks identified by the teams are not purely technical but procedural and relational, linked to who controls data, how risk is communicated, and which institutions are empowered to act.

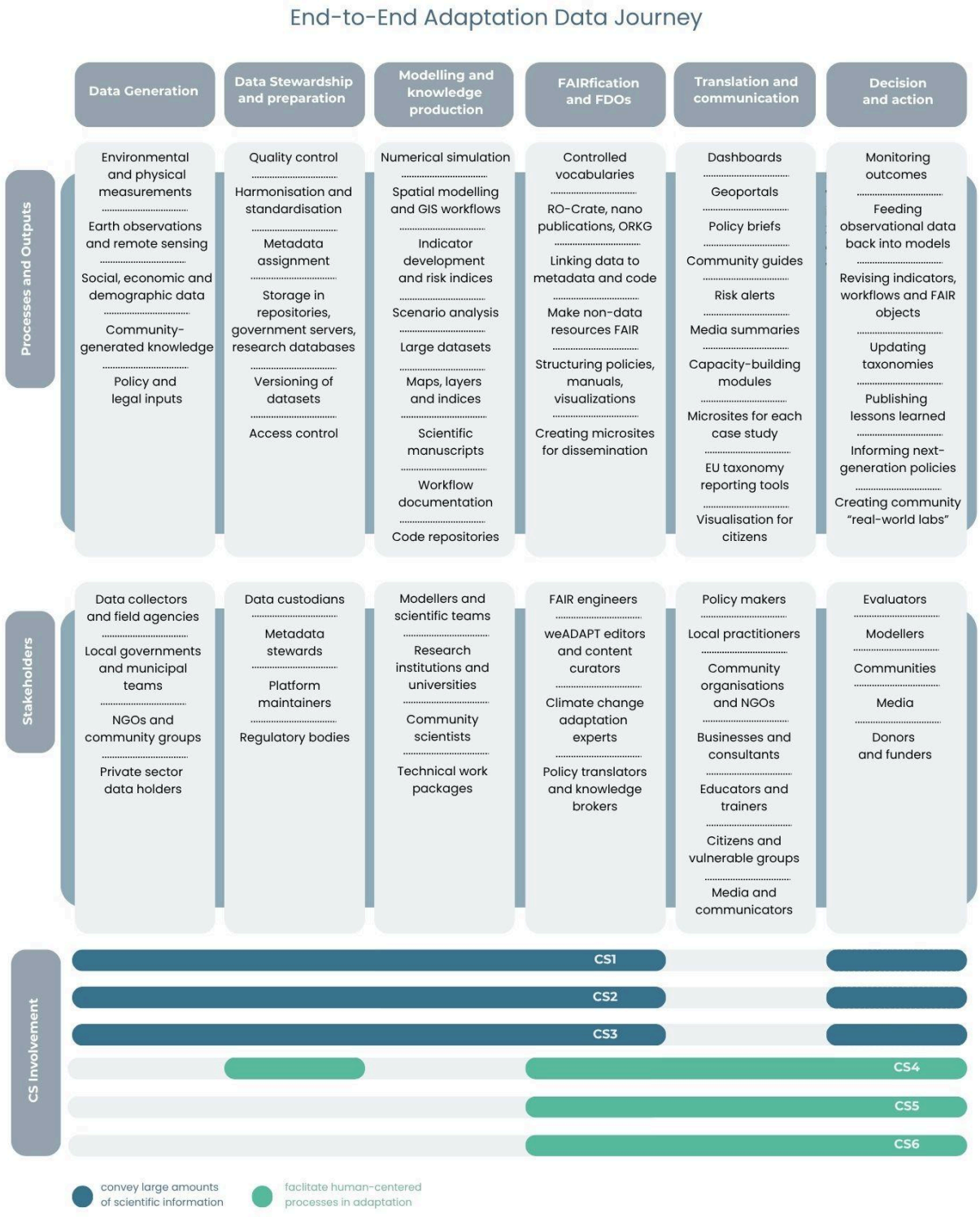


Figure 1. End-to-End Adaptation Data Journey. Illustrating the stages through which climate adaptation knowledge moves, from generation to decision, including the main processes, stakeholders, and the involvement of each case study. This framework guided the analysis presented in Sections 3.1–3.6.

In this sense, FAIR2Adapt's contribution extends beyond the technical implementation of the FAIR principles. The project is showing how FAIR thinking can serve as an institutional language, a way to bridge disciplines, sectors, and epistemic cultures. The Hamburg case, for instance, demonstrates how long-term collaboration between data custodians and municipal planners can lead to the internalization of FAIR workflows within city processes. The RiOMar and Arctic cases, meanwhile, reveal how FAIR approaches can support interoperability across environmental data infrastructures that already function at continental or global scales. Conversely, the CCH and weADAPT cases illustrate how FAIRness can operate at the opposite end of the spectrum, by connecting policy makers, grassroots, qualitative, and narrative knowledge in ways that preserve context while making it discoverable. Figure 2 illustrates the possible synergies and complementarities between case studies.

An important cross-cutting finding is that FAIRification processes must match the rhythm of adaptation. Technical harmonization cannot occur in isolation from the cycles of engagement, trust-building, and validation that make data credible to users. Case studies that began from established relationships, such as Hamburg, advanced more quickly in turning data into action because trust already existed. Those working across multiple institutions or countries, such as RiOMar or the Arctic case, faced longer feedback loops and had to balance scientific precision with accessibility. The CCH, positioned as a bridge, underscores how semantic tools like its underlying FAIR Climate Connectivity vocabulary can help shorten these loops by creating a common foundation of interoperability across cases.

The variation in engagement maturity across case studies reflects structural differences in how each entered the project. Case Study 3 (City of Hamburg), Case Study 4 (FCID), Case Study 5 (SEI-Oxford) and Case Study 6 (ADELPHI) built on pre-existing relationships with climate change adaptation practitioners and decision-makers, enabling substantive co-design activities within Year 1. Case Study 1 (NERSC) and Case Study 2 (IFREMER), while having fewer established pathways to end-user communities, bring complementary strengths. Case Study 1 combines scientific modelling expertise with strong data management capabilities and FAIR knowledge, positioning it to demonstrate how the FAIRification framework can be applied to complex climate model outputs. This makes Case Study 1 a valuable reference case for understanding what support climate change adaptation communities need to adopt FAIR practices. Case Study 2, meanwhile, will apply the design thinking methodology used in other case studies to strengthen its stakeholder engagement. Sections 3.1.4 and 3.2.4 describe how each case study will build on these foundations in Year 2.

The expected impact of these findings is threefold:

- First, within the consortium, the lessons from each case study inform the design of shared FDOs and data workflows that respond to real end-user needs rather than abstract compliance checklists. The CCT, for example, is emerging as a reference model that can anchor other cases' outputs semantically.

- Second, for external adaptation communities, the work highlights replicable methods for linking science, governance, and practice, from Arctic monitoring to coastal management and urban risk planning.
- And third, at the European and global scale, the project contributes to the ongoing debate on what “FAIR for adaptation” truly means. It moves the concept beyond data repositories and into the lived processes of decision-making, policy, and capacity building.

Taken together, the six case studies demonstrate that the FAIR principles are not just technical standards but social contracts: agreements about how information should move between people and institutions in the service of resilience. By uncovering the frictions, dependencies, and innovations that shape this movement, FAIR2Adapt is helping to redefine the pathways through which climate knowledge becomes action.

Additionally, these case studies also illustrate how technical solutions aiming at end-to-end frameworks are bounded by human processes, and that often the scope of what is needed for adaptation is not readily available from FAIRification processes. These six studies stand as testimony that implementation processes also face important trade-offs where adaptation needs have to be balanced and prioritized, not only maximizing the impact of the information but also understanding that limitations exist in what is possible to deliver from a technological perspective. This is an important dimension to take into account as a learning for this consortium and other data-driven adaptation efforts.

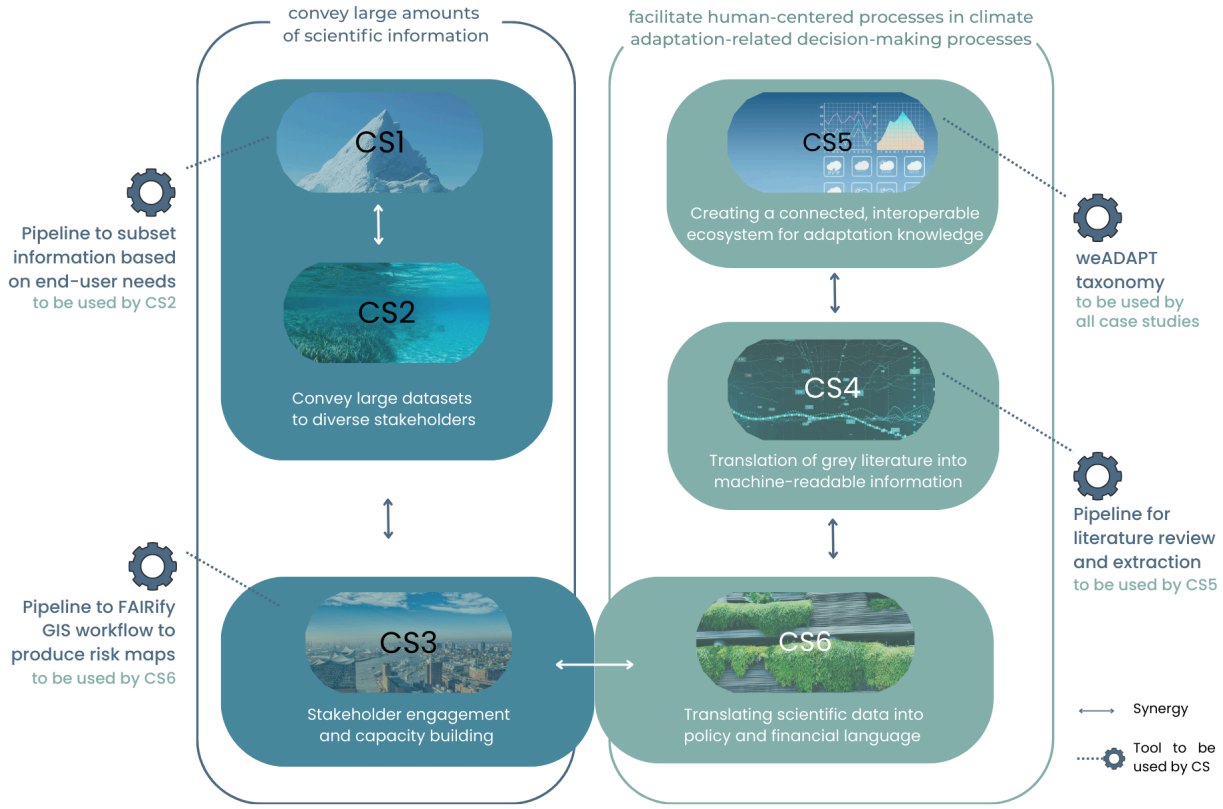


Figure 2. - Synergies and Complementarities Between Case Studies showing how CS1–CS3 contribute to scientific data production and modelling, while CS4–CS6 strengthen downstream processes such as knowledge translation, capacity building and decision support. Arrows indicate shared pipelines, needs or tools across the consortium.

5. Next Steps

Maintaining engagement throughout the next phases of FAIR2Adapt requires balancing two complementary forms of collaboration: one centred on the technical integration of data and standards across work packages, and another rooted in the sustained dialogue with stakeholders who represent the ultimate users of the project's outputs. Both dimensions are equally important. The technical progress achieved through FAIRification must remain grounded in the real decision-making needs and languages of practitioners, policymakers, and communities. Likewise, engagement processes must continue to generate feedback that directly informs how tools and frameworks evolve.

Maintaining engagement will build on the networks and working relationships established during the first year of the project. Each case study has developed its own rhythm and structure for collaboration, ranging from long-term institutional partnerships, as in Hamburg and Portugal, to wider ecosystems of practitioners and communities, as in weADAPT. Over the next phase, these different engagement modalities will be coordinated into a structure of recurring exchanges between case study leads and technical work packages. This will include periodic check-ins with WPs 2, 3, and 4 to align the development of FAIR vocabularies, taxonomies, and implementation profiles; cross-case reflection sessions facilitated by WP6 to identify synergies and shared bottlenecks; and targeted workshops focused on testing prototypes, dashboards, and semantic structures with end users.

To ensure continuity, engagement will not end with consultations. Each cycle of interaction will include explicit feedback loops which serve as mechanisms for testing, reflection, and iteration. In practice, this means that draft versions of FAIR vocabularies, metadata profiles, and user interfaces will be shared with case study partners for early validation, followed by revisions that reflect their input. The role of weADAPT will be particularly important in this phase, serving as a living interface where the results of different cases can be made visible, tested, and discussed in community spaces that extend beyond the consortium itself.

Translating knowledge needs into actionable outputs will continue to rely on the design-thinking approach used throughout the first project year. The workshops revealed that many information needs are not linear or isolated but interdependent, requiring joint reflection between data producers, intermediaries, and users. The next step is to convert these needs into concrete FDOs, semantic tools, and decision-support materials. For example, the taxonomy being further developed through the CCH and I-ADOPT will underpin the tagging and interlinking of datasets from the Arctic, RiOMar, and Hamburg cases, enabling their reuse in new decision contexts. Similarly, the FIPs being refined by WPs 2 and 3 will provide templates for how to publish, share, and monitor adaptation knowledge consistently across systems.

Each of these outputs will be oriented toward the moment of decision - where data becomes action. Some will take the form of dashboards and visualization tools tailored to decision-makers, while others will focus on standardizing metadata and licensing so that information can circulate seamlessly between platforms. A key emphasis will be on documenting and measuring how FAIRification enhances usability, transparency, and impact (D6.2), ensuring that lessons are transferable to future adaptation initiatives.

Continuity with upcoming deliverables will be ensured through deliberate alignment with the project's technical and synthesis outputs. Deliverable 5.1 focuses on the creation of FAIR processes and implementation profiles, which are directly informed by the user needs and stakeholder insights captured in D6.1. Similarly, upcoming deliverables under WP2 and WP3 will operationalize the taxonomies, vocabularies, and metadata structures that connect the technical backend of FAIR2Adapt with the applied use cases described here. Later in the

project, Deliverable 6.2 will return to the case studies to assess how these tools have been applied, providing evidence of impact and lessons for scaling.

In short, engagement will remain an iterative and adaptive process, anchored in the dialogue between the human and the technical. FAIR2Adapt's next phase will transform the needs, stories, and challenges identified so far into tangible artifacts that others can use, modify, and build upon. Through this ongoing loop of co-creation and feedback, the project will continue to translate knowledge into action and action back into learning, building not only interoperable data systems but also a more connected and resilient community of practice for climate adaptation.

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6. Annex I - Methodological Background

1. Overview

This annex provides the conceptual and methodological background that informed the design and facilitation of the FAIR2Adapt workshops. It complements the condensed methodology section in the main document by explaining the theoretical foundations of design thinking, its relevance for adaptation research, and the rationale behind the workshop structure used across the six case studies.

The content in this annex is not essential for understanding the findings of Deliverable 6.1, but it offers deeper insight into why specific methods were chosen, how they connect to FAIRification processes, and how they support user-centred knowledge elicitation.

2. Design Thinking as an Approach for Adaptation Knowledge Elicitation

Design thinking is a user-centred methodology developed to support the rapid development and refinement of products, services, and systems in contexts where multiple stakeholder groups interact and where needs are often dynamic or not fully articulated. It has been widely applied in innovation settings (Brown, 2009; Liedtka, 2015; Giacomin, 2014), and increasingly in environmental research and data-intensive projects, due to its ability to reveal latent needs and system bottlenecks.

Design thinking is particularly appropriate for climate adaptation and FAIRification processes because:

- adaptation involves complex, multi-actor systems with heterogeneous priorities
- stakeholders often struggle to articulate their information needs in conventional technical terms
- data infrastructures and decision-making processes must evolve iteratively
- early identification of usability constraints can prevent misalignment between technical outputs and real-world needs

In FAIR2Adapt, design thinking offered a structured yet flexible framework for exploring how adaptation knowledge is produced, interpreted, and acted upon across different environmental and institutional settings.

3. Canonical Phases of Design Thinking

While design thinking is often presented as a sequence of five phases, these phases are typically iterative and overlapping:

1. **Empathise** — Understand users' experiences, motivations, and pain points.
2. **Define** — Identify core needs, bottlenecks, and problem statements.
3. **Ideate** — Generate options and creative solutions.
4. **Prototype** — Draft simplified versions of products, interfaces, or processes.
5. **Test** — Evaluate prototypes with users and refine based on feedback.

In practice, design thinking rarely follows a linear path. Instead, teams move back and forth between phases as new insights emerge.

FAIR2Adapt applied primarily the *empathise* and *define* phases—eliciting stakeholder needs, mapping decision processes, and identifying bottlenecks. The *ideate*, *prototype*, and *test* phases are planned to occur later in the project through the development of FAIR Digital Objects (FDOs), vocabularies, dashboards, and data-access tools.

4. Role of Agile Ethnographic Methods

The empathise phase often draws on agile ethnographic techniques, which help uncover:

- tacit knowledge
- emotional or behavioural drivers
- informal workflows
- unspoken constraints in decision-making
- mismatches between formal and lived processes

In FAIR2Adapt, exercises based on Jungian archetypes, projective imagery, and narrative prompts were used to help participants articulate needs that are difficult to express in purely technical terms. This proved especially useful for:

- revealing institutional dependencies
- understanding perceived versus actual data flows
- identifying which stakeholders influence decisions informally
- surfacing latent tensions or gaps in communication

These insights informed the mapping of stakeholder ecosystems and data journeys for each case study.

5. Why Design Thinking Was Chosen for FAIRification Work

The FAIR principles aim to make data Findable, Accessible, Interoperable, and Reusable. However, FAIRification is not only a technical exercise: it requires understanding the needs of human actors who must produce, curate, interpret, and apply data.

Design thinking was selected because:

- FAIR processes require alignment between technical infrastructures and user behaviour
- stakeholder needs often exceed or diverge from what technology can deliver
- iterative refinement helps prevent “solutionism” by grounding development in practical constraints
- qualitative adaptation knowledge (policies, narratives, decisions) is difficult to structure without user input
- FAIR Digital Objects must support real workflows, not abstract idealisations

This approach helped clarify where FAIRification can meaningfully support adaptation and where institutional or cultural factors limit its uptake.

6. Partial Implementation in FAIR2Adapt

Design thinking can be used fully (from empathise to test) or partially, depending on project needs. FAIR2Adapt applied a **partial implementation**, focusing on:

- eliciting user needs
- identifying information gaps
- mapping processes
- clarifying expectations for data products
- prioritising which FAIRification pathways warrant development

7. Relationship to Other Work Packages

The insights generated through design-thinking exercises fed directly into:

- **WP5**: requirements for FAIR Implementation Profiles
- **WP3**: development of FAIRification pipelines
- **WP2**: metadata, vocabularies, and technical foundations
- **WP6**: synthesis of adaptation knowledge needs

By grounding technical development in stakeholder realities, this approach ensures that FAIR2Adapt’s outputs can be meaningfully integrated into adaptation decision-making contexts.

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7. Annex II - Case Study Supplementary Material

This annex brings together all detailed stakeholder mappings and supplementary visual materials that provide detailed information about the descriptions of the case study sections in the main document. It includes the full tables of actors, roles, needs, and challenges identified during the workshops, as well as the accompanying images such as stakeholder maps, process diagrams, brainstorming outputs, and early dashboard mockups. Together, these tables and figures offer a granular view of the diverse ecosystems in which each case study operates, illustrating how data providers, policymakers, technical teams, community groups, and other actors interact within real-world adaptation processes.

Actor	Examples of actors	Processes / Needs / Challenges
Knowledge custodians	Met institutes, NERSC modellers, dose-sensor networks, Copernicus Arctic MFC	Provide large amounts of complex scientific information in a format that is understandable by different stakeholders at both long times and short term.
Platform owners	Crisis Committee for Nuclear and Radiological Preparedness, Norwegian Institute for Water and the Environment (NIVA), EBAS Thredds Data Frameworks for information	Real-time Access Protocols, Receiving data, Compiling datasets, Comparing information, usage metrics
End-users	County governors, fisheries & aquaculture, Small Scale Nuclear Reactor vendors, media, residents	Subset datasets, tailored briefs and reports, embeddable charts

Table A1. Main stakeholder roles identified in the workshop of Case Study 1.

Actor	Examples of actors	Processes / Needs / Challenges
Data Provider	Watershed Agency	Agency provides information to RIOMar.
End user from Industry	Aquaculture professionals	Very focused on a very small part of the project. Aren't convinced that this will serve their needs.
End user from Science and Policy - Biodiversity Expert	Marine Protected Area Manger	The project has focused on biochemistry. They do not feel the project serves their needs either.
End user - Scientists and Data Analysts	Research institutions using this dataset	Using the data for the past 20 years. Critical of the quality.
Facilitator and Connector	French Biodiversity Office	Researchers in charge of applying environmental policy. Helpful in redefining outputs. Facilitate the interface between researchers, water agencies and professionals.
Researcher	External researchers working on the CC effects on river fluxes and on the future of nutrient inputs in the rivers	Not a high priority stakeholder. They are willing to explain the outputs of their past projects.
Project Lead and Knowledge broker	Ifremer DevOps	Understands how all stakeholders can interact to improve the decisionmaking process based on information.

Table A2. Main stakeholder roles identified in the workshop of Case Study 2.

Actors	Examples of actors	Processes / Needs / Challenges
Data Providers	LGV - City Agency for Geoinformation	Information is handled correctly
End user - City budget / administrative workflow	Senate	Budget prioritization Impact Evaluation Information
End User - Adaptation and sustainability coordinators	Ministry for Environment, Climate, Energy and Agriculture, Climate Adaptation Coordination Unit	Reliable information Actionable Insights Compare Layers easily Easily define and share potential sites and interventions
Data Providers - Knowledge providers: Government Scientists	Water Agency scientists	Provide accurate data and expert opinion Participate in articulating the end-to-end process of models to policy.
End user - Civil Society	Citizens	Easily digestible information
Researchers	Professors, Researchers, Students	Build risk models
Data custodians, metadata stewards	Ministry for Environment, Climate, Energy and Agriculture, Climate Adaptation Coordination Unit LGV - City Agency for Geoinformation	Provide accurate and reliable information that is useful

Table A3. Main stakeholder roles identified in the workshop of Case Study 3.

Type	Geographical scale	Name	General description
Knowledge platform	European	weADAPT	Collaborative platform for CCA.
Knowledge platform Data portal	Global	Climate Risk Dashboard	Database explorer of future climate change impacts and (un)avoidable risks from cities to the global scale.
Knowledge platform	European	Climate-ADAPT	Platform to support European countries in adapting to climate change through sharing information on risks, vulnerabilities, adaptation strategies and EU policies, and adaptation solutions.
Knowledge platform	European	MIP4Adapt	Platform to support European regional and local authorities to prepare and plan their adaptation plans.
Data portal	Global	IIASA SSP Scenario Explorer 3.1.0	Climate scenario explorer with projections for national demographic and economic models for each SSPs.
Data portal	National	Portal do Clima	Platform that provides climate and weather information, offering data, forecasts and analyses, for monitoring climate trends.
Data portal	Global	Copernicus Climate Change Service C3S	EU thematic information service about climate. Provides high-quality data, tools and analyses to support climate monitoring.
Knowledge platform Data portal	National	Agência Portuguesa do Ambiente (APA)	Platform for accessing environmental data and information managed by Portugal's national environmental authority.
Community network	National	Adapt.local	Network that connects municipalities and partner organizations to support local CCA. Provides resources, trainings and promotes peer learning.
Strategies for adaptation to climate change	National and Regional	Strategy of Lisbon Metropolitan Area	Provides key principles, tools and practical approaches to help governments, organizations and communities adapt to climate change.
Research articles	Global	Examples: Scopus, ISI Web of Knowledge	Peer-reviewed publications that provide data, methodologies or reviews that contribute to advancing knowledge and informing policy.
Other types of grey literature	Global	-	Publications such as reports, policy briefs and technical documents that provide information outside formal academic publishing.

Table A4. Selected examples of resources potentially relevant for CCA communities in Portugal.

Actor	Examples of Actors	Processes, Needs and Challenges
Data producers and providers	Researchers from public institutions and private companies	**Process:** Create datasets, conduct studies and produce information that can be used by other actors and create knowledge. **Need:** Know the requirements and gaps of information from knowledge creators and practitioners to produce meaningful data. **Challenge:** Data producers are easily disconnected from end-users and unaware of their needs.
Knowledge producers and End-users - Business and private companies	Consultants, companies, advisors	**Process:**Try to find ways to adapt assets and processes to CCA, while navigating policy, regulatory frameworks and market drivers (e.g. ESG and mitigation). **Need/challenge:** Policy and regulatory frameworks are often published on non-user friendly databases (e.g. Diário da República) using formats that enable machine readability (e.g. computer scan of official documents).
Data producers, Knowledge producers - NGOs	Quercus, ZERO	**Process:** Produce studies, communication elements and engage with science-policy-society interface, trying to push for climate action while raising awareness. **Need/challenge:** The lack of up to date and meaningful data represents a bottleneck on knowledge production.
Knowledge producers and custodians - Governmental agencies	Portuguese Environmental Agency (APA)	**Process:** Responsible for the studies and policy recommendations that support the National and Municipal executives (Ministry of Environment and Municipalities) in its CCA decision-making processes and policy development. **Need/challenge:** FAIR data management can improve the applicability and usability of the data (and knowledge).
Knowledge custodians	Research groups, IPMA, Copernicus	**Process:** Intimately related to data providers, but have the complementary role of sharing the curated data through an interface, working as brokers at a more technical level and could cover from globe to local scale. **Need/challenge:** FAIR data management can improve the applicability and usability of the data (and knowledge).

Actor	Examples of Actors	Processes, Needs and Challenges
Policy owners - National level executives	Sectoral Ministries	**Process:** Responsible for mainstreaming CCA into sector specific policy processes (e.g. agriculture, economy). **Need/Challenge:** They receive up to date and reliable information provided by knowledge producers and, with the support of experts, incorporate new knowledge to improve existing policies.
Knowledge brokers - Networks of adaptation practitioners	Adapt.local, Lisbon Metropolitan Area	**Process:** Responsible for connecting and encourage the relationship between data providers, knowledge producers and end-users. **Need/Challenge:** Improve local adaptation policies and practises at the regional and local scale.
Knowledge brokers - Municipal executives	Executives at national (Ministry of Environment) and local levels (municipalities and similar regional and local governmental agencies)	**Process:** Responsible for the national or local level CCA decision-making processes, policy proposal and adoption. **Need:** They need knowledge that is easily interpreted and, if needed, processed by different sets of tools. **Challenge:** Data is often not packaged for easy analysis and is not curated for use by executive-level adaptation practitioners.
End-users - Adaptation practitioners, Business and private companies.	Practitioners at all scales (municipalities and other regional public agencies) Adaptation practitioners and planners	**Process:** Must plan for adaptation, using scientific data and local knowledge. **Need/Challenge**: Adaptation planning includes navigating complex administrative and policy processes including budgetary constraints (in the case of public agencies).
End-users - Court of auditors	Tribunal de Contas	**Process:** Responsible for auditing the application of CCA policies and measures (e.g. strategies, plans and practical activities). **Need/Challenge**: Must have access to the latest policy documents at all levels (national, regional and local).
End-users - Media and Communication Channels	National networks, newspapers, blogs	**Process:** Aim at communicating CCA information to society, by convening scientific knowledge and policy discourses, while providing room for balanced discussion, raising awareness and science-policy-society interfaces.

Actor	Examples of Actors	Processes, Needs and Challenges
		Need/Challenge: To ensure the release of an accurate and liable 'story', media should be granted access to the latest and most updated information and knowledge.
End-users - Local communities, civil society and civil society organizations	Citizens	**Process:** Citizens who want to take informed actions, be inspired, and create movement in the form of small local initiatives. **Need/Challenge**: Require specific and clear examples of CCA solutions, often not easily accessible.

Table A5. Selected examples of stakeholder roles identified in the workshop of Case Study 4.

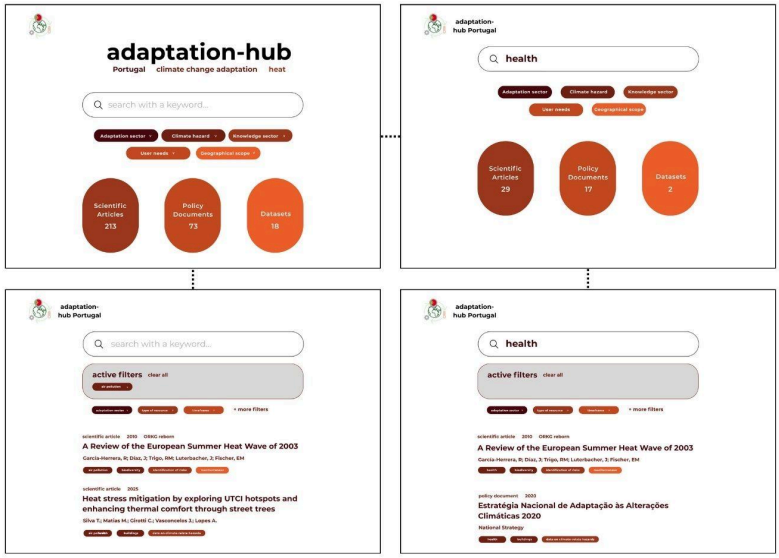


Figure A1. First proposed mockup of the dashboard and hub of Case Study 4.

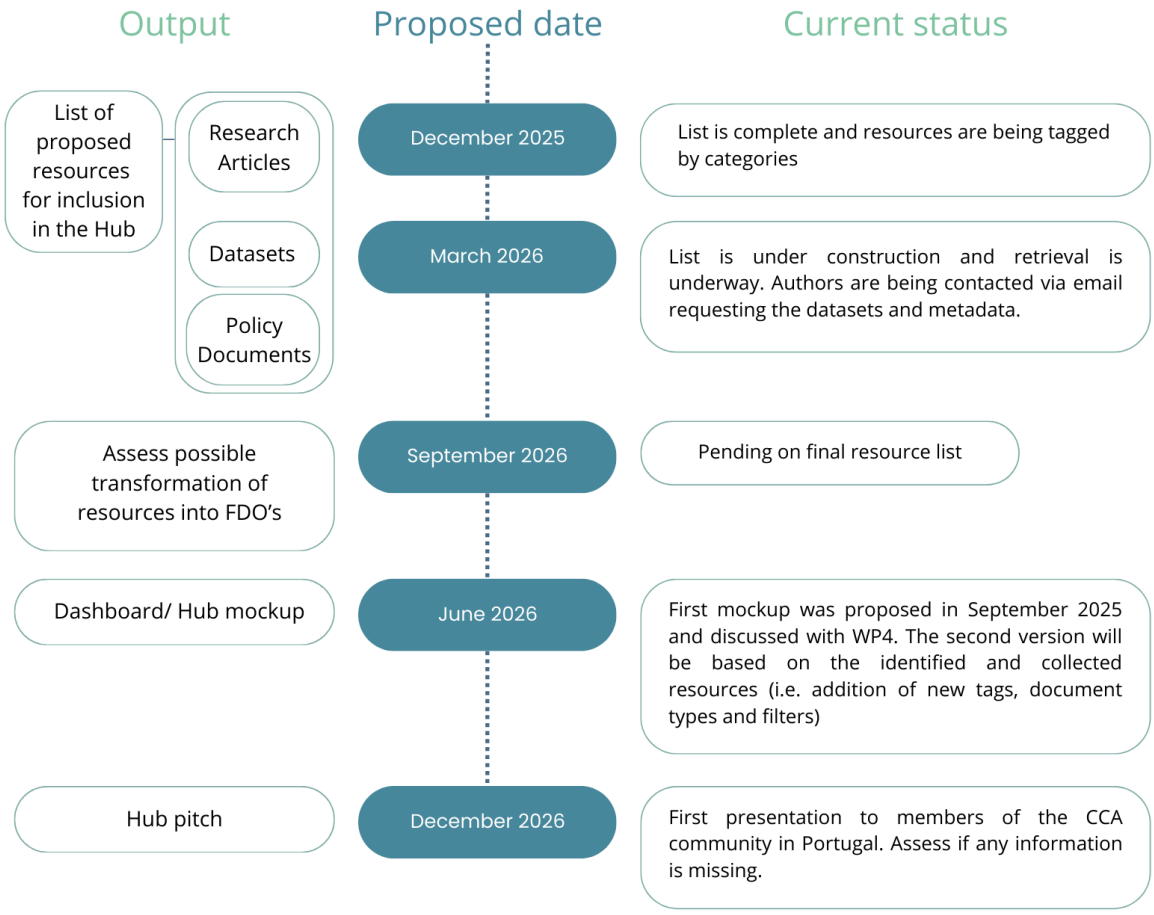


Figure A2. Timeline and outputs of Case Study 4.

Actor	Examples of Actors	Role / Needs
End Users – Policy Makers and Decision Makers	Municipal, national and international policy officers; planners; government departments; ministries	Use weADAPT and related tools to inform funding and prioritisation of adaptation measures. Require concise, credible, and ready-to-use information. Struggle with information overload and need short cost–benefit briefs and visual summaries that support quick, defensible decisions.

Actor	Examples of Actors	Role / Needs
Knowledge Brokers – NGOs and Community Organisations	Local NGOs, international humanitarian organisations, civil society groups	Support vulnerable communities and facilitate knowledge exchange but often lack time and resources. Need simplified templates to share data, access to peer networks, and capacity-building materials to translate complex information into action.
Knowledge Custodians – Editors and Platform Managers	weADAPT editors, SEI Oxford staff, partner institutions managing sub-portals	Ensure content is reliable and verified. Require clear quality-control mechanisms, comparison tools, and licence checks to manage the growing volume of contributions and maintain FAIR compliance.
Funders and Sponsors	Development agencies, philanthropic foundations, multilateral donors	Provide financial support for research and knowledge platforms. Need clarity about impact, regional needs, and funding overlaps to avoid duplication. Seek robust evidence of project effectiveness and value.
Knowledge Producers – Researchers and Academics	Universities, research centres, and scientific institutions contributing studies and data	Produce evidence and models that inform adaptation strategies. Need support in communicating their findings in accessible, policy-relevant formats. Struggle to bridge the gap between scientific publication and practical use.
Capacity Developers and Educators	Trainers, teachers, professional networks, and adaptation course developers	Design and deliver training materials based on adaptation knowledge. Require modular lesson plans, clear learning pathways, and tracking tools to monitor participation and learning outcomes.
End Users – Local Communities and Community Groups	Local adaptation groups, municipal councils, residents in vulnerable areas	Experience direct climate impacts and need practical, locally relevant knowledge. Require plain-language summaries, translation into local languages, and visibility for their lived experiences in the global knowledge space.

Actor	Examples of Actors	Role / Needs
Intermediaries – Local Practitioners and Emergency Responders	Municipal technicians, local planners, civil defence staff	Implement adaptation and response measures on the ground. Need quick-access checklists, simple hazard maps, and SMS or mobile tools that are usable in real-time situations.
Knowledge Users – Citizens	Individuals seeking to understand local risks	Experience direct climate impacts and need practical, locally relevant knowledge. Require easy-to-understand explanations of complex risks, visual tools, and opportunities for participation and sharing local stories.
Data Custodians – Private Sector and Insurers	Water utilities, transport companies, insurance providers	Hold critical datasets for adaptation planning but face regulatory and commercial barriers to sharing. Need frameworks for data exchange with appropriate recognition and data-protection safeguards.
Technical Contributors – Modellers and Developers	Research institutions, data scientists, GIS specialists	Develop and maintain model codebases and datasets. Need clear provenance systems, integration with open repositories, and mechanisms to publish model outputs transparently.
Facilitators – Community Champions and Innovators	Local leaders, activists, educators, citizen scientists	Translate global knowledge into local context and share local experiences upwards. Need visibility, recognition, and tools to showcase successful examples of community-led adaptation.
Public Bodies	Environmental agencies, local authorities, ministries	Produce, compile, and communicate official adaptation data while being accountable to the public. Need clear dashboards, performance indicators, and secure collaboration tools.
Knowledge Producers – Conservation and Environmental Organisations	Environmental NGOs, charities, field research organisations	Conduct their own studies on biodiversity and ecosystem restoration. Need comparative insights, tools to share data, and easy upload formats for reports and geo-referenced observations.

Actor	Examples of Actors	Role / Needs
Knowledge Holders – Indigenous and Hard-to-Reach Communities	Indigenous groups, rural associations, remote communities	Often excluded from digital information networks. Need offline resources, culturally appropriate tools, and recognition of oral knowledge traditions.

Table A6. Main stakeholder roles identified in the workshop of Case Study 5.

Actor	Examples of Actors	Role / Needs
Strategic Coordinator	Department 43 (Referat 43)	**Role:** Central steering actor coordinating diverse processes and mediating between multiple institutional interests. **Needs:** Broad stakeholder engagement; alignment of priorities; clear data flows. **Challenges:** High coordination burden; dependency on external data owners; risk that its strategic role becomes invisible amid competing institutional agendas.
Professional Associations	Medical Association, Pharmacists' Association	**Role:** Provide professional expertise and support the implementation of climate-related measures. **Needs:** Clear, concrete information on risks and measures. **Challenges:** Limited staff and time may reduce continuity of engagement.
Neighbourhood Advisory Boards	Local quarter advisory boards (Ortsteilbeiräte)	**Role:** Direct interface with residents; facilitate democratic participation in decision-making at neighbourhood level. **Needs:** Clear information about local climate risks, city actions, and citizen engagement pathways. **Challenges:** Variable accessibility; independence can create communication gaps; hard to integrate into citywide processes.

Actor	Examples of Actors	Role / Needs
Infrastructure and Utilities Providers	SWB (utilities)	**Role:** Key providers and managers of essential infrastructure data. **Needs:** Integrated systems and shared data protocols. **Challenges:** Restrictive data policies; siloed datasets; inconsistent cooperation depending on staff; limited integration with GeoInfo systems.
Citizens	Residents, vulnerable groups, local initiatives	**Role:** Both recipients of adaptation measures and active co-creators through behaviour change and local participation. **Needs:** Clear, simple information on risks, municipal actions, and how to contribute. **Challenges:** Engagement varies by perceived vulnerability; low transparency reduces trust and participation.
NGOs and Civil Society Organisations	Local environmental groups, social NGOs, advocacy coalitions	**Role:** Catalysts for community engagement, providing expertise and social mobilisation. **Needs:** Transparent and timely information on adaptation efforts, policy changes, and new data. **Challenges:** Loss of involvement or lack of transparency can quickly erode trust and create resistance.
Geospatial Data Authorities	GeoInfo (city geoinformation office)	**Role:** Collect, manage, and provide spatial data for planning and adaptation monitoring. **Needs:** Better coordination mechanisms; greater visibility and use of datasets. **Challenges:** Missing datasets can become bottlenecks; lack of coordination leads to integration costs and delays.

Actor	Examples of Actors	Role / Needs
Data Controllers	Data stewards and managers in other departments	**Role:** Prepare and provide department-specific data; interface with cross-departmental information flows. **Needs:** Clear priorities, interoperability processes, manageable data protection constraints. **Challenges:** Complex cooperation across silos; misaligned priorities; data protection officers may limit data sharing.
Political Decision Makers	Elected officials, city assembly members	**Role:** Set priorities and authorise climate adaptation measures. **Needs:** Visible, defensible evidence for decisions; clear communication of adaptation benefits. **Challenges:** Changing political priorities or budget constraints introduce uncertainty; low transparency undermines continuity.
Municipal Implementing Authorities	Authorities in Bremerhaven	**Role:** Implement measures derived from state-level initiatives; key in ensuring local uptake. **Needs:** Equal partnership; meaningful participation in planning; clear communication. **Challenges:** Low involvement reduces effectiveness; partnership essential for impact.
Health Sector Authorities	Health Authority	**Role:** Lead on health-related adaptation (e.g., heat risks). **Needs:** Accessible, actionable information on health impacts. **Challenges:** Heavy workloads and competing priorities can reduce attention to adaptation issues.

Actor	Examples of Actors	Role / Needs
Heritage and Building Protection Authorities	Monument Protection Authority	**Role:** Evaluate and approve adaptation measures affecting protected buildings. **Needs:** Alignment between adaptation options and heritage requirements. **Challenges:** Strict preservation rules can limit possible measures; review processes may delay implementation.

Table A7. Main stakeholder roles identified in the workshop of Case Study 6.

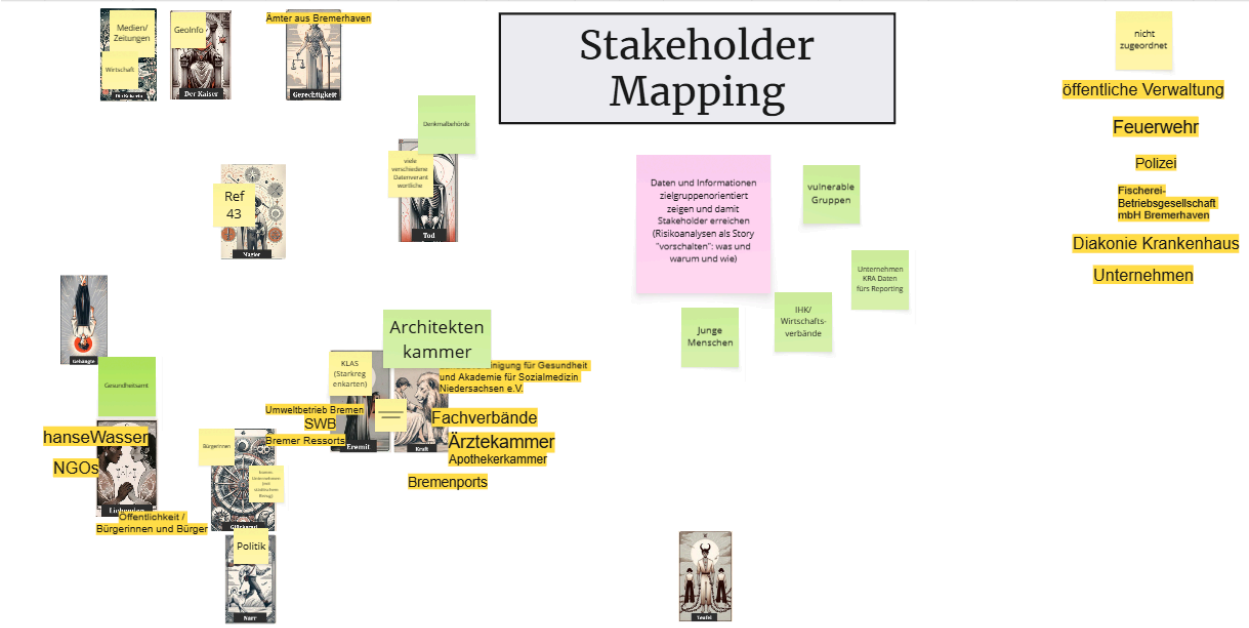


Figure A3. Stakeholder mapping exercise of Case Study 6.

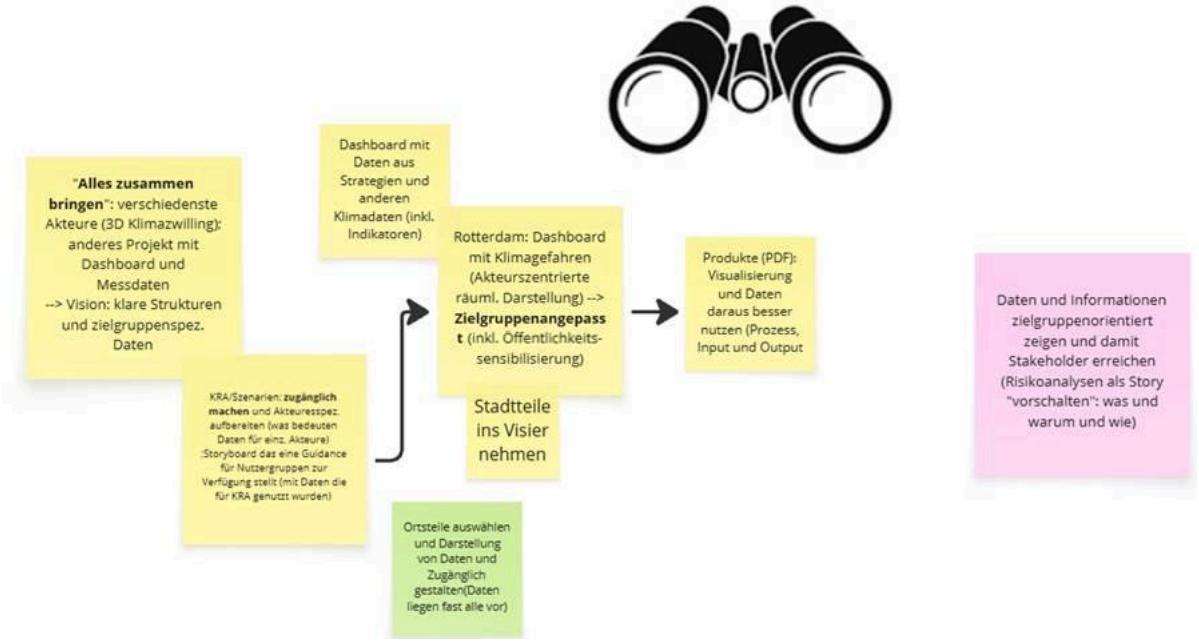


Figure A4. Brainstorming workshop on vision and goals of Case Study 6.

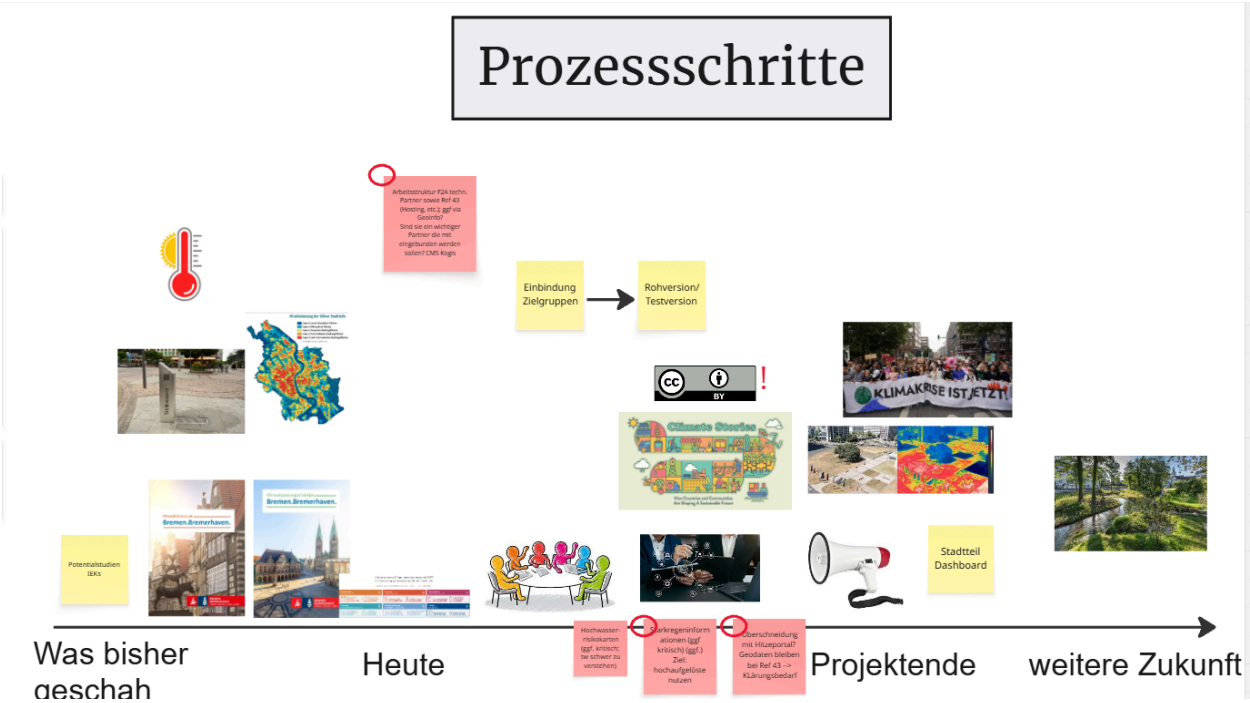


Figure A5. Process Mapping Exercise of Case Study 6.

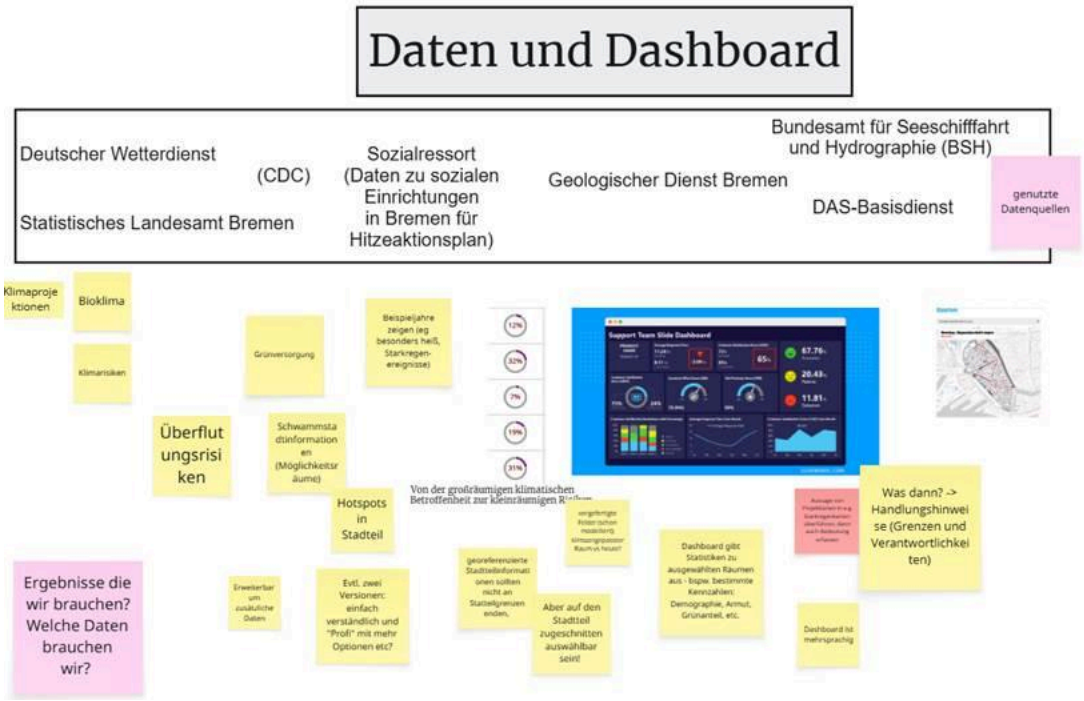


Figure A6. Workshop brainstorming on dashboard prototype of Case Study 6.